One Way or Another: Modes of Transport and International Trade

Edoardo Tolva *

ISEG Research and Universidade de Lisboa

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Abstract

The transport sector is the backbone of international trade yet has faced major disruptions in recent years. This paper studies how substitution across transport modes shapes trade and welfare. I develop a quantitative multicountry trade model with multiple transport modes and endogenous congestion costs. Exploiting the closure of Russian airspace as a quasi-natural experiment, I estimate an average elasticity of substitution between air and sea transport of 2.6. I also provide the first estimates of congestion effects in air cargo. Counterfactual simulations show that mode substitution attenuates welfare losses from mode-specific transport shocks.

Keywords: Transport, Substitution, Ukraine, Congestion, Trade costs

JEL Classification: F10, F12, F14, F18, R4

*Edoardo Tolva: ISEG Research, Universidade de Lisboa, Rua do Quelhas 6, 1200-781 Lisbon, email: edoardo.tolva@iseg.ulisboa.pt. I am very grateful to Dennis Novy and Marta Santamaria for their constant guidance and support in this project. I want to thank for the useful comments: Facundo Albornoz, Prottoy Akbar, Tibor Besedes, Natalie Chen, Paola Conconi, Davide Del Prete, Banu Demir, Ana Fernandes, Alejandro Graziano, Anna Ignatenko, Pamina Koenig, Fabrizio Leone, Thierry Mayer, Gianmarco Ottaviano, Stephen Redding, Thomas Sampson, Bengt Söderlund, Carmen Villa, and Maurizio Zanardi. I also thank conferences and seminars participants at the Warwick Macro & International Workshop, CEP/Warwick Junior Trade Workshop, GEP-CEPR Annual Postgraduate Conference, RIEF, EAYE, European Urban Economic Association, SNEE, ETSG, Royal Economic Society Summer School, Journal of International Economics Summer School, Norwegian School of Economics, and Napoli Parthenope. All errors are my own.

1 Introduction

Transport plays a pivotal role in international trade, enabling the global movement of goods and sustaining both consumption and production across countries. In recent years, the transport sector has been subject to unprecedented disruptions arising from both geopolitical events and natural phenomena. The Russian invasion of Ukraine in 2022 led to the closure of Russian airspace, severely disrupting air traffic between Europe and Asia, a corridor accounting for more than 20% of global air cargo volume (IATA, 2022). As air cargo has become increasingly important for international trade (Hummels, 2007; Feyrer, 2019), these restrictions carry potentially large economic consequences. Similarly, in 2023 and 2024 maritime trade was disrupted by events affecting the Panama Canal and the Red Sea, two critical chokepoints for global shipping.[1]

A common feature of these events is that they affect one transport mode at a time. The closure of Russian airspace primarily disrupted air routes, while attacks in the Red Sea targeted maritime transport. As a result, unaffected modes become relatively cheaper, creating incentives for substitution across transport modes that may partially offset the negative impact on trade flows. Yet, much of the international trade literature abstracts from this margin of adjustment by assuming that agents rely on a single transport mode. This assumption not only ignores heterogeneity across modes but also precludes substitution in response to mode-specific trade cost shocks.

Motivated by the central role of transport in global trade and the recent sequence of mode-specific disruptions, this paper studies how substitution across transport modes can act as a buffer against increases in transport costs.

I address this question using both empirical and theoretical approaches. First, I document the widespread use of multiple transport modes within narrowly defined product categories and origin–destination pairs. Second, I develop a quantitative multi-sector Ricardian model of trade that incorporates multiple transport modes and endogenous transport costs. Third, I exploit an exogenous trade cost shock to provide novel estimates of the elasticity of substitution between transport modes. Finally, I use the model to evaluate a set of counterfactual scenarios in which a specific transport mode is disrupted, quantifying the implications for welfare.

While maritime transport continues to dominate global trade by volume, air cargo has grown steadily in both volume and value (Hummels, 2007; Feyrer, 2019; IATA, 2025). Using Eurostat

[1] Around 30,000 ships per year transit these two waterways, accounting for more than 20% of global seaborne trade. Source: Unctad (2024).

data, I show that air transport accounts for approximately 30% of extra-European trade value and that more than 70% of origin–destination–sector triplets use both air and sea transport. This evidence indicates that agents engaged in long-distance trade typically have access to multiple transport modes and can therefore adjust their mode choice when one becomes more costly.

Motivated by these facts, I extend a standard multi-sector Ricardian model of trade (Eaton and Kortum, 2002; Dekle et al., 2007; Caliendo and Parro, 2015) to allow for multiple transport modes in international shipping. The choice of transport mode is modeled as a discrete decision made after imports are determined for a given origin. Importers select the mode based on transport prices and idiosyncratic preferences for shipments between two locations. This structure delivers two key equilibrium objects: the mode-specific trade shares and a transport cost price index. The latter can be interpreted as an iceberg trade cost that, together with country-specific price distributions, determines aggregate trade flows. A central parameter of the model is the elasticity of substitution between transport modes, which is specific to each sector and governs the extent to which agents can reallocate shipments across modes. A key advantage of this framework is that it nests within the "universal gravity" structure (Allen et al., 2020), allowing standard quantitative trade models to be extended in a tractable way to incorporate multiple transport modes.

Allowing for substitution across transport modes introduces a novel margin of adjustment relative to the standard trade literature. This margin has two main implications. First, welfare losses from increases in transport costs are mitigated as agents partially switch to alternative modes. Second, the change in the transport cost index varies across country pairs depending on their exposure to the disrupted mode, generating asymmetric effects that cannot be replicated by models with a single iceberg trade cost.

The model also features congestion forces. As demand for a given transport mode rises, its price increases, making bilateral transport costs endogenous to usage. Consequently, a mode-specific cost shock triggers two opposing effects. On the one hand, substitution toward the unaffected mode raises its price through congestion. On the other hand, higher transport costs reduce bilateral trade volumes, which alleviates congestion and partially offsets the initial shock. By solving the model in changes, I show that the only additional data requirement relative to a standard Ricardian model is information on initial trade shares by mode for each country pair.

I estimate the elasticity of substitution between transport modes using the Ukraine–Russia conflict as an exogenous shock to transport costs. Sanctions imposed between the European Union and Russia led to the closure of Russian airspace, significantly increasing air transport costs (IATA, 2022; Chu et al., 2024). I implement a difference-in-differences (DiD) strategy

that compares changes in flight times and trade shares by mode between affected and unaffected country pairs before and after the outbreak of the war. To this end, I combine flight-level data from nearly two million flights recorded by OpenSky with Eurostat data on trade flows. I first show that the conflict substantially increased flight times between Europe and East Asia, and then document a corresponding shift toward maritime transport in affected routes. Dividing the estimated effect on mode shares by the effect on transport times yields a Wald-DiD estimate of the elasticity of substitution between modes (De Chaisemartin and d'Haultfoeuille, 2023). I estimate this elasticity to be approximately 2.6. I also document substantial heterogeneity across sectors and products: goods that are non-perishable and have a high value-to-weight ratio, such as textiles, exhibit among the highest substitutability across modes. The implied sectoral elasticities range from about 1.6 for metal products and machinery to above 4 for textiles and motor vehicles.

As an alternative identification strategy, I exploit the 2021 disruption at U.S. West Coast ports (Hu et al., 2021). A key advantage of this setting is that I directly observe transport charges paid by importers. Using a similar DiD design, I define treated countries as U.S. trading partners in Asia and use European countries as a control group. Both empirical strategies yield statistically similar estimates of the elasticity of substitution between transport modes. Given the prolonged nature of these disruptions, I interpret the estimated elasticities as reflecting medium- to long-run adjustments.

In addition to the substitution elasticity, the model features a congestion parameter governing how transport prices respond to usage. I estimate this parameter using a shift-share instrumental variables approach (Borusyak et al., 2022, 2025). Using U.S. Census data on imports, I construct instruments based on variation in imports across origins within narrowly defined sectors.[2] As a robustness check, I also use European exports to non-U.S. destinations as an alternative shifter. I estimate the congestion parameter to be approximately 0.06 for maritime transport, consistent with existing estimates such as Fuchs and Wong (2024), and around 0.10 for air transport. To my knowledge, this paper provides the first estimate of congestion effects in air cargo transport.

I calibrate a multi-sector version of the model, which follows Caliendo and Parro (2015) without input-output linkages across sectors, to 2019 data on bilateral trade from BACI, trade by mode of transport from UNCTAD, and production from UNIDO. The calibration covers 16 manufacturing sectors, two transport modes, air and sea, and 50 entities: 47 economies, a European bloc, a China with Hong Kong and Macao, and the rest of the world. Trade elasticities

[2]Eurostat does not report transport costs, which prevents estimating congestion effects using European data.

are sector specific and taken from Caliendo and Parro (2015), and the elasticities of substitution between modes are the sectoral estimates described above. I use the calibrated model to simulate the effects of transport disruptions on trade flows and welfare.

I conduct two counterfactual exercises: the closure of Russian airspace and the closure of the Suez Canal. I compare a baseline model with endogenous mode substitution to a model with fixed Cobb–Douglas shares across modes and to a model without transport modes. In both cases, allowing for substitution reduces welfare losses, as agents partially reallocate shipments toward unaffected modes: average losses are 3.4 percent larger in the Suez Canal experiment, and 6.4 percent larger for the entities directly affected by the airspace closure, when modal shares are held fixed. Ignoring transport modes altogether overstates losses by considerably more, by 24 percent in the Suez Canal experiment, because the shock then applies to entire bilateral flows rather than to the share carried by the affected mode. While these aggregate effects may appear modest, even small welfare changes can have substantial economic significance at the level of entire economies.

From a policy perspective, the results highlight the importance of accounting for transport mode substitution when designing trade policies. Mode-specific disruptions, whether caused by geopolitical events, infrastructure bottlenecks, natural phenomena, or regulations, do not translate one-for-one into trade cost increases once agents can reoptimize their transport choices. Ignoring this adjustment margin can lead to systematic overestimation of welfare losses from transport shocks and mismeasurement of their distribution across countries. These findings suggest that policies aimed at improving the resilience of international trade, such as infrastructure investments or mode-specific regulations, should be evaluated in a framework that jointly considers multiple transport modes, congestion effects, and endogenous substitution. The framework also lends itself to environmental applications, such as evaluating policies that target the emissions of individual transport modes.[3]

Related Literature. This paper contributes to the literature studying the relationship between international trade and transport modes. Existing work typically frames this relationship as a trade-off between speed and cost, emphasizing the value of timely delivery in hedging against uncertainty or responding to demand shocks. The choice of transport mode has been shown to depend on the quality and characteristics of shipped goods (Hummels and Skiba, 2004; Hummels, 2007; Hummels and Schaur, 2013), delivery timing and reliability (Evans and Harrigan,

[3] A previous version of this work also included an application to enviromental policy which is now developed as a separate project.

2005; Hummels and Schaur, 2010), and relative prices across modes (Micco and Serebrisky, 2006; Harrigan, 2010). Technological progress has played a central role in the long-run expansion of air transport in international trade (Campante and Yanagizawa-Drott, 2018; Feyrer, 2019). Even when studying transport disruptions, however, the literature has largely focused on the direct effects on the affected mode, abstracting from the possibility that firms adjust by switching to alternative modes (Carballo et al., 2014; Söderlund, 2020; Coşar and Thomas, 2021; Feyrer, 2021; Sandkamp et al., 2022; Al-Malk et al., 2024; Besedeš et al., 2024; Ludwig, 2025).

From an empirical perspective, this paper provides novel estimates of the elasticity of substitution between transport modes at the country-pair level. To my knowledge, this is the first study to identify this elasticity using quasi-natural experiments that generate plausibly exogenous variation in relative transport costs. I also estimate the elasticity separately by sector. These estimates reveal substantial heterogeneity across goods and discipline the sector-specific elasticities in the quantitative model, so that the response of modal shares to a shock depends on the sectoral composition of each country pair's trade. Hummels and Schaur (2013) provides early evidence of substitution across modes and reports elasticities of comparable magnitude. More recently, Fuchs and Wong (2024) estimates a related parameter for road and rail transport, while Ko et al. (2025) recovers sectoral trade elasticities by mode exploiting geographic variation across country pairs.[4]

From a theoretical perspective, the paper augments standard multicountry Ricardian models of trade (Eaton and Kortum, 2002; Caliendo and Parro, 2015; Allen et al., 2020) by allowing for multiple transport modes, thereby enabling the analysis of mode-specific trade cost shocks and their aggregate welfare consequences. In doing so, it contributes to the growing literature on endogenous transportation costs in international trade and spatial economics (Allen and Arkolakis, 2014; Asturias, 2020; Brancaccio et al., 2020; Fajgelbaum and Schaal, 2020; Wong, 2022; Jaworski et al., 2023; Fuchs and Wong, 2024; Do et al., 2024). This paper is the first to combine substitution across transport modes with endogenous transportation costs driven by congestion and demand in a tractable multicountry general equilibrium framework. In addition, I provide the first estimates of the congestion elasticity for air transport, using a shift-share identification strategy (Borusyak et al., 2022, 2025), and show that its magnitude is comparable to estimates for other transport modes (Fuchs and Wong, 2024).

[4] An earlier unpublished working paper by Lux (2011) also examines substitution across transport modes in international trade. That study differs substantially from this paper. Empirically, it relies on descriptive correlations rather than quasi-exogenous variation. Theoretically, it does not provide a structural framework linking observed mode shares to a well-defined elasticity of substitution.

2 Data and Facts

In this section, I describe the main dataset used in the empirical analysis and present summary statistics on trade by transport mode that motivate the model's structure.

2.1 Eurostat Data

I use monthly data from Eurostat on European imports and exports by transport mode for trade with non-European countries. Trade flows are reported both in value (euros) and weight (kilograms).[5] The data distinguish four main transport modes—air, rail, road, and sea, which together account for approximately 95% of total trade in the sample.[6] Flows are reported at the 6-digit Harmonized System (HS) product level, allowing for a detailed analysis of transport mode choice. The sample covers the period from January 2010 to December 2022.

I focus on trade flows shipped by air and sea, which together account for roughly 80% of total trade value, in order to avoid measurement issues associated with land transport. A limitation of the Eurostat data is that, for some country pairs, trade is occasionally reported using transport modes that are geographically implausible. For example, imports from the United States to Germany are sometimes recorded as arriving by road or rail. This occurs because customs declarations are based on the point of entry into the Eurozone rather than the original mode of international shipment.[7]

To reduce the prevalence of zero trade flows, I restrict the sample to the 80 largest countries by GDP (World Bank, 2019), which together account for approximately 93% of global trade. I also exclude the United Kingdom, which ceased reporting to Eurostat after 2020, as well as Russia and Ukraine due to the ongoing conflict.

The Eurostat dataset has two main limitations. First, it does not report transport costs by mode. Second, it only covers extra-European trade.[8] Although intra-EU trade flows are available from 2010 onward, they are reported at a more aggregated level using the Standard Goods Classification for Transport Statistics (NST), which consists of 20 sectors and is collected through surveys rather than customs declarations. Moreover, most intra-EU trade is conducted by road (Santamaría et al., 2023), reflecting shorter distances and geographic constraints that

[5] All transactions with a reported weight below 100 kg are recorded as zero in Eurostat.

[6] I exclude the remaining modes, such as unknown, post, fixed mechanisms, inland waterways, and self-propulsion, either because they are rarely used or cannot be consistently classified.

[7] For instance, a shipment flown from the United States to Switzerland and subsequently transported by truck to Germany without passing through Swiss customs is recorded as a German import from the United States by road. Such reporting issues are common in international trade data (Cotterlaz and Vicard, 2023).

[8] For example, Germany–China trade is included, whereas Germany–France trade is not.

make air and maritime transport less relevant within Europe.

2.2 U.S. Census Bureau Data

As an alternative data source, I also use data from the U.S. Census Bureau,[9] which provide monthly information on U.S. trade by transport mode through an API. As in the Eurostat data, trade flows are reported at the 6-digit HS level. The sample covers the period from January 2013 to December 2022.

In contrast to Eurostat, the U.S. Census data report only import flows and distinguish only between air and sea transport. This reflects the geography of the United States, which shares land borders exclusively with Canada and Mexico. A key advantage of this dataset is that it reports total freight charges by origin–sector–mode, allowing me to directly observe transport costs rather than relying on proxies.

2.3 Composition of European Imports and Exports by Transport Mode

This section documents two stylized facts about transport modes in European trade with non-European partners. First, air transport accounts for a substantial share of trade value alongside sea transport. Second, the two modes coexist within narrowly defined origin–destination–product triplets. Together, these facts motivate a Ricardian model of trade with multiple transport modes.

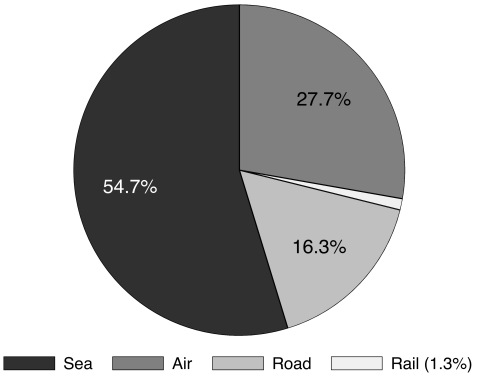
Fact I: Relevance of Multiple Transport Modes. Figure 1a reports the composition of extra-European trade by transport mode in 2019. Despite its higher cost, air transport carries approximately 760 billion euros of goods, about 28% of total extra-European trade value. The pattern is very similar for imports and exports separately (Figure H.1), and air and sea transport together account for almost all trade with China and the United States, two of Europe's largest trading partners (Figure H.2).

Mode shares vary considerably across ISIC3 sections, but air transport is common in many of them (Figure 1b). Air transport dominates in goods with high value-to-weight ratios, such as electronics, pharmaceuticals, and precision machinery, whereas sea transport dominates in heavy, low-value goods such as raw materials and petroleum products (Hummels, 2007). Since air and sea transport account for the vast majority of extra-European trade, I focus on these two modes for the remainder of the analysis.

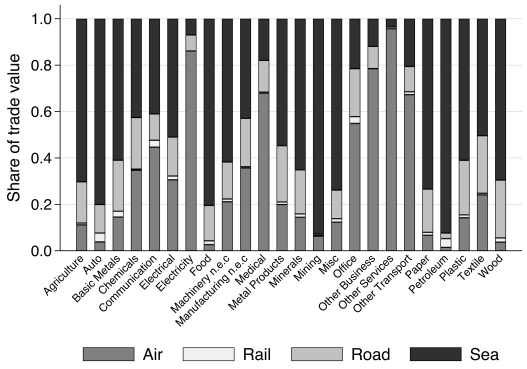
[9] See, for example, Hummels (2007); Hummels and Schaur (2010, 2013).

Figure 1: Transport Mode Share of Trade

(a) Aggregate



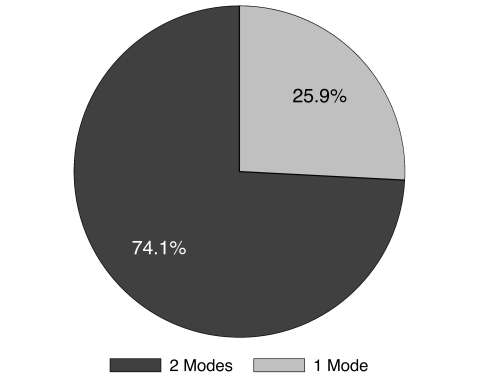
(b) ISIC3 Sections



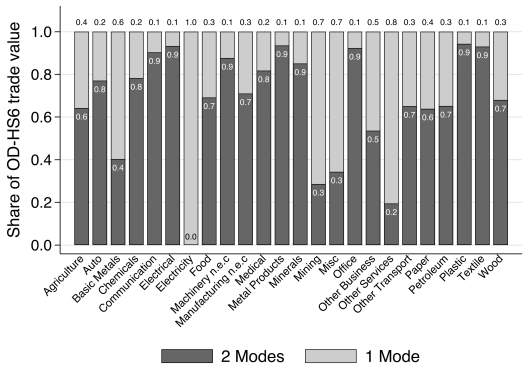
Notes: Panel (a) reports the aggregate share of trade value in euros by transport mode. Panel (b) reports the same statistics by ISIC3 section. The sample consists of imports and exports of European Union members in 2019, collected from Eurostat.

Figure 2: Number of Modes Used per Origin–Destination–HS6

(a) Aggregate



(b) ISIC3 Sections



Notes: Panel (a) shows the share of trade value accounted for by origin–destination–HS6 triplets that use both air and sea transport. Panel (b) shows the same statistic by ISIC3 section. The sample consists of imports and exports of European Union members in 2019, collected from Eurostat.

Fact II: Multiple Modes within the Same Origin–Destination–Product. The sectoral pattern above suggests that product characteristics determine the mode of transport. If they did so entirely, each origin–destination–product triplet would ship through a single mode. Instead, triplets that use both air and sea transport account for approximately 74% of trade value (Figure 2, panel (a)).[10] This holds for imports and exports separately (Figure H.4) and is even more pronounced for trade with the United States and China (Figure H.5). Moreover, multimodal shipping is prevalent across all ISIC3 sections (Figure 2, panel (b)). The coexistence of the two modes is therefore a pervasive feature of international trade, not an artifact of a narrow set of products.

Taken together, these facts show that multiple transport modes matter both in the aggregate

[10]The pattern is robust to using unweighted data or to weighting trade flows by kilograms rather than values (Figure H.3).

and within narrowly defined products. U.S. Census Bureau data display very similar patterns in trade value and weight by mode (Appendix D.1), consistent with earlier evidence from disaggregated U.S. data (Hummels and Schaur, 2013). Trade models that assume a single transport mode therefore abstract from an empirically important margin. The next section incorporates multiple transport modes into a simple Ricardian framework.

3 The Model

I combine a discrete choice model of transport modes with a multi-sector Ricardian trade framework. The model builds on a standard multi-country trade model (Eaton and Kortum, 2002; Caliendo and Parro, 2015), which determines global trade flows, wages, and prices. Two salient features are added to this framework. First, a discrete choice model over transport modes endogenously determines the equilibrium modal shares in bilateral trade and delivers a transport price index, which enters the trade model as an iceberg trade cost. In addition, not all sectors are assumed to have multiple modes available.[11] Second, the model features congestion costs at the origin–destination–mode level that affect transportation prices along each route.

3.1 Setup and Demand

The economy consists of N countries and $\mathcal{J}$, indexed by $j \in \mathcal{J}$, tradable sectors partitioned into $\mathcal{J}_M$ and $\mathcal{J}_S$. Single-mode sectors are shipped by a single transport mode only, while sectors $\mathcal{J}_M$ allows for multiple modes of transport. Within each sector j there is a continuum of goods $u \in [0,1]$ produced under perfect competition. Country n is endowed with L_n units of labor, supplied inelastically and freely mobile across sectors within a country but immobile across countries. Each country has a non-tradable sector whose output enters final consumption through a Cobb–Douglas aggregator with country-specific share $1-\alpha_n$.

Upper-tier preferences over all tradable-sector composites and the non-tradable good are Cobb–Douglas:

$$U_n = \left(\prod_{j \in \mathcal{J}} \left(Q_n^j\right)^{\alpha_n^j} \right) \left(Q_n^{NT}\right)^{1-\alpha_n}, \tag{1}$$

where Q_n^j denotes the CES composite of sector-j goods with within-sector elasticity σ, Q_n^{NT} is the non-tradable composite, and $\alpha_n^j > 0$ is the share of total absorption X_n devoted to final

[11] Empirical evidence on the distribution of modal shares by sector is discussed in Section 2 and illustrated in Figure 2

demand in sector j. The sector shares satisfy

$$\sum_{j \in \mathcal{J}} \alpha_n^j = \alpha_n \qquad \forall\, n, \tag{2}$$

so the non-tradable expenditure share is $1 - \alpha_n$. Country n devotes

$$F_n^j = \alpha_n^j X_n \tag{3}$$

to final expenditure on sector j. Following Dekle et al. (2007), the exogenous aggregate trade deficit is $D_n = X_n - Y_n$, and the sectoral trade deficit is

$$D_n^j \equiv X_n^j - Y_n^j, \tag{4}$$

where $D_n^j > 0$ if country n is a net importer in sector j; these satisfy $\sum_{j \in \mathcal{J}} D_n^j = D_n$.

3.2 Transport Costs

The bilateral transport-cost index d_{ni}^j is the only dimension along which $\mathcal{J}_M$ and $\mathcal{J}_S$ sectors differ. All other equilibrium objects, such as input-bundle costs, price indices, trade shares, market clearing, and welfare, depend on sector type solely through d_{ni}^j.

3.2.1 Multimodal Sectors

For $j \in \mathcal{J}_M$, goods can be shipped by any of $M_j \geq 2$ transport modes. I assume a two-step decision process: importers first choose the source country and then select the transport mode. Factory-gate prices are independent of transport mode: producers set prices solely as a function of their productivity, while transport costs, borne by the importer, do not affect this pricing decision.[12] All transport modes are simultaneously available, and the transport sector operates under perfect competition.[13]

Conditional on choosing origin i, importers in country n select among M_j modes. Following Anderson et al. (1987), a CES demand system for transport can be derived from a nested logit model with a deterministic second stage, the derivation is given in Section E.1 of the

[12] This assumption is standard in trade models and allows trade costs to be expressed as a fraction of the good's value.

[13] Empirical evidence shows that many goods are shipped using both air and sea transport, as illustrated in Figure 2.

Supplementary Appendix . The aggregate transport-cost index for sector-j trade from i to n is

$$d_{ni}^{j} = \left(\sum_{m=1}^{M_j} \left(d_{ni}^{j,m} \right)^{1-\eta_j} \right)^{1/(1-\eta_j)} \qquad \forall j \in \mathcal{J}_M, \tag{5}$$

which aggregates mode-specific costs into a single iceberg cost index, with d_{ni}^{j} playing the role of the standard iceberg trade cost in each sector. The share of sector-j bilateral trade carried by mode m is

$$\pi_{ni}^{j,m} = \frac{\left(d_{ni}^{j,m} \right)^{1-\eta_j}}{\sum_{k=1}^{M_j} \left(d_{ni}^{j,k} \right)^{1-\eta_j}} = \frac{X_{ni}^{j,m}}{X_{ni}^{j}}, \tag{6}$$

where $X_{ni}^{j,m}$ is the value of sector-j imports by country n from country i shipped by mode m, X_{ni}^{j} is the corresponding total bilateral import value in sector j, and $\eta_j > 1$ is the sector-specific elasticity of substitution across transport modes. The additional condition $\eta_j > \sigma$ is required for the mode-choice problem to be well posed conditional on the source country.

3.2.2 Single-Mode Sectors

For $j \in \mathcal{J}_S$, all goods in the sector are shipped by a single transport mode, which I denote m_j. In this case there is no choice of the transport mode so the model boils down to a standard trade model. The transport costs is therefore just a function of the iceberg cost of the available mode:

$$d_{ni}^{j} = d_{ni}^{j,m_j} \quad \forall\, j \in \mathcal{J}_S, \tag{7}$$

and consequently the share by mode in this case is equal to 1.

3.2.3 Elasticity of Substitution

The sector-specific elasticity η_j, for $j \in \mathcal{J}_M$, is the new key parameter of this framework. Taking logs of equation (6) and taking ratios between two generic modes m and s within sector j yields the estimating equation

$$\ln\left(\frac{X_{ni}^{j,m}}{X_{ni}^{j,s}} \right) = (1-\eta_j) \ln\left(\frac{d_{ni}^{j,m}}{d_{ni}^{j,s}} \right), \tag{8}$$

The coefficient $(1-\eta_j)$ captures the elasticity of the modal trade share with respect to changes in relative transport costs. Its magnitude can vary systematically across sectors, reflecting the

fact that goods and products intrinsic characteristics, such as weight or temperature sensitivity, may affect their degree of substitutability.

3.3 Congestion

I assume mode-specific transport costs, $d_{ni}^{j,m}$, between countries i and n. I further assume that these costs depend on a fundamental component, $\delta_{ni}^{j,m}$, and a congestion component, $\Xi_{ni}^{j,m}$. Congestion costs apply for all sectors, $j \in \mathcal{J}$.

I model $\Xi_{ni}^{j,m}$ as consisting of two elements. The first is a function of the total value of trade shipped by mode m between i and n in sector j, which I write as $(X_{ni}^{j,m})^{\lambda_m}$, with $0 < \lambda_m < 1$. The parameter λ_m measures the sensitivity of congestion costs to trade volumes. A higher λ_m implies that trade costs respond more strongly to changes in bilateral trade flows by mode. The second element is a pair-specific component that can be interpreted as the capacity, or quality, of the infrastructure used to ship goods, which I define as $I_{ni} = I_n \times I_i$. This approach is similar to that used in recent spatial models (Fajgelbaum and Schaal, 2020; Allen and Arkolakis, 2022; Fuchs and Wong, 2024). The total costs of moving a good with mode m from country i to country n are then given by:

$$d_{ni}^{j,m} = \delta_{ni}^{j,m} \Xi_{ni}^{j,m} = \delta_{ni}^{j,m} \frac{\left(X_{ni}^{j,m}\right)^{\lambda_m}}{I_n I_i}, \tag{9}$$

where $\delta_{ni}^{j,m}$ denotes the fundamental iceberg trade cost for mode m between i and n in sector j, and $d_{ni}^{j,m}$ is the total cost of shipping goods using mode m. The mode-specific iceberg costs, $\delta_{ni}^{j,m}$, capture all pair-specific determinants of transport costs other than infrastructure and congestion and are assumed to be exogenous to the economy.

Since individual importers cannot influence market prices, I treat congestion costs as externalities that importers take as given when choosing the transport mode. In Appendix E.1, I derive the expression equivalent to equation (6). For the rest of the paper, I assume that the capacity and quality of infrastructure are constant across countries and modes.[14]

[14] Endogenizing infrastructure investment would substantially increase the model's complexity and would require a dynamic setting that is outside the scope of this work. At the end of this section, I show that, under this assumption, the level of infrastructure is not needed to compute counterfactuals in changes.

3.4 Equilibrium in Changes and Welfare

The remainder of the model follows a standard Ricardian trade framework based on Caliendo and Parro (2015) with the only difference that I do not assume input-output linkages.[15] More details about production, trade flows, and market clearing conditions can be found in Section F of the Supplementary Appendix .

I apply the exact hat-algebra approach of Dekle et al. (2007) defining $\hat{x} = x'/x$ for any variable x, and holding $\{A_i^j,\ \theta_j,\ \eta_j,\ L_n,\ I_n,\ D_n,\ D_n^j\}$ fixed. Where x' is the post-shock equilibrium value. The change-form system follows from applying hat algebra sector by sector to the level-form equations of Section F, with parameters indexed by j throughout.

The change in the sectoral trade share is

$$\hat{\pi}_{ni}^j = \frac{\left(\hat{d}_{ni}^j\,\hat{c}_i^j\right)^{-\theta_j}}{\sum_{h=1}^{N} \pi_{nh}^j \left(\hat{d}_{nh}^j\,\hat{c}_h^j\right)^{-\theta_j}}, \tag{10}$$

and the change in the sectoral price index is

$$\hat{P}_n^j = \left[\sum_{i=1}^{N} \pi_{ni}^j \left(\hat{d}_{ni}^j\,\hat{c}_i^j\right)^{-\theta_j}\right]^{-1/\theta_j}, \tag{11}$$

where $\hat{c}_i^j = \hat{w}_i^{\beta_i^j}\,\hat{P}_i^{j\,(1-\beta_i^j)}$. These hold identically across both sector types; the partition affects only how $\hat{d}_{ni}^j$ is constructed. For $j \in \mathcal{J}_M$, the change in the aggregate transportation cost, $\hat{d}_{ni}$, reflects both the direct change in costs for the affected mode and the induced price responses of other modes through congestion effects. In Appendix E.3, I show that the resulting impact on the transport cost index from a change $\hat{\delta}_{ni}^s$ can be written as:

$$\hat{d}_{ni}^j = \left[1 + \left((\hat{\delta}_{ni}^{j,s}(\hat{X}_{ni}^{j,s})^{\lambda_s}\right)^{1-\eta_j} - 1\right)\pi_{ni}^{j,s} + \sum^{-s} \pi_{ni}^{j,m}\left((\hat{X}_{ni}^{j,m})^{\lambda_m(1-\eta_j)} - 1\right)\right]^{1/(1-\eta_j)}, \tag{12}$$

This formulation has two key features. First, it requires very little additional data relative to a standard quantitative trade model, namely the share of trade by mode between country pairs. Second, it implies that changes in aggregate trade costs differ across country pairs even when mode-specific cost changes are identical across routes. This occurs because the overall effect depends on the pre-shock modal shares. In this sense, these shares can be interpreted as mea-

[15] Including a richer production structure tends to amplify the shock transmission so that any effect in this model will be amplified once i-o linkages are considered.

sures of exposure to mode-specific shocks. Intuitively, the more intensively a mode is used along a given route, the more exposed that country pair is to increases in transport costs for that mode. These heterogeneous effects may amplify cross-country differences. For example, developing countries rely more heavily on sea transport, so an increase in sea transport costs would disproportionately affect economies that are already more vulnerable.

For the sectors in which we have a single mode, $j \in \mathcal{J}_S$, instead we have that:

$$\hat{d}^j_{ni} = \hat{\delta}^{j,m_j}_{ni} (\hat{X}^j_{ni})^{\lambda_{m_j}}. \tag{13}$$

Applying hat algebra to the budget constraint (39) of Section F sector by sector and summing across $\mathcal{J}_M \cup \mathcal{J}_S$, with gross output expressed as $Y^{j\prime}_n = (\beta^j_n)^{-1}(\alpha^j_n X'_n - D^j_n)$ in the post-shock equilibrium, yields the change-form budget constraint:

$$\hat{w}_n Y_n + D_n - \sum_{j=1}^{J} \frac{D^j_n}{\beta^j_n} = \sum_{j=1}^{J} \sum_{i=1}^{N} \frac{\pi^j_{ni} \left(\hat{d}^j_{ni} \hat{c}^j_i \right)^{-\theta_j}}{\sum_h \pi^j_{nh} \left(\hat{d}^j_{nh} \hat{c}^j_h \right)^{-\theta_j}} \left[\frac{\alpha^j_i}{\beta^j_i} (\hat{w}_i Y_i + D_i) - \frac{1 - \beta^j_i}{\beta^j_i} D^j_i \right], \tag{14}$$

where $Y_n = w_n L_n$ is pre-shock GDP and j sums over $\mathcal{J}_M \cup \mathcal{J}_S$; the deficits $\{D_n, D^j_n\}$ are retained in levels as exogenous objects.

Welfare. With Cobb–Douglas upper-tier preferences (1) and own-sector roundabout production (36), the ACR result (Arkolakis et al., 2012) applies with trade elasticity θ_j and roundabout amplification $1/\beta^j_n$ for country n in sector j. The welfare change for country n is

$$\widehat{W}_n = \prod_{j \in \mathcal{J}_M \cup \mathcal{J}_S} \left(\hat{\pi}^j_{nn} \right)^{-\alpha^j_n / (\theta_j \beta^j_n)}. \tag{15}$$

The formula holds uniformly across both sector types; the partition affects only how $\hat{\pi}^j_{nn}$ responds to a given shock through the type-specific construction of $\hat{d}^j_{ni}$.

4 Estimation

In this section, I estimate the elasticity of substitution between transport modes, η, the key parameter of the model, together with the congestion parameter, λ. Identification of η relies on plausibly exogenous variation in mode-specific transport costs generated by large geopolitical and logistical disruptions. I exploit the closure of Russian and Ukrainian airspace in 2022, which increased air transport costs on Europe–Asia routes without directly affecting maritime

transport. Because Eurostat data do not report transport prices, I complement this strategy by estimating the congestion parameter, λ, separately using a shift-share approach based on U.S. data.

4.1 Substitution Between Transport Modes

The Russian invasion of Ukraine in February 2022 led to a sharp deterioration in diplomatic and economic relations between Russia and several sanctioning countries, among which is the European Union.[16] One immediate consequence was the mutual closure of airspace between the opposing blocs, while Ukrainian airspace was closed to civilian flights for security reasons. As a result, European carriers were forced to reroute flights to Asian destinations, leading to substantial increases in flight times.[17]

The Russian airspace is a critical corridor for Europe–Asia trade: approximately 36 % of European extra-regional trade involves Asian partners.[18] The closure of Russian and Ukrainian airspace therefore generated a large and asymmetric shock to international transport networks. Crucially, this event affected only one transport mode at a time. By increasing flight distances and travel times, the closure raised air transport costs without directly affecting maritime shipping. This asymmetry provides clean variation in relative mode-specific transport costs and allows the elasticity of substitution to be identified from changes in relative mode shares within narrowly defined country pairs and product categories.

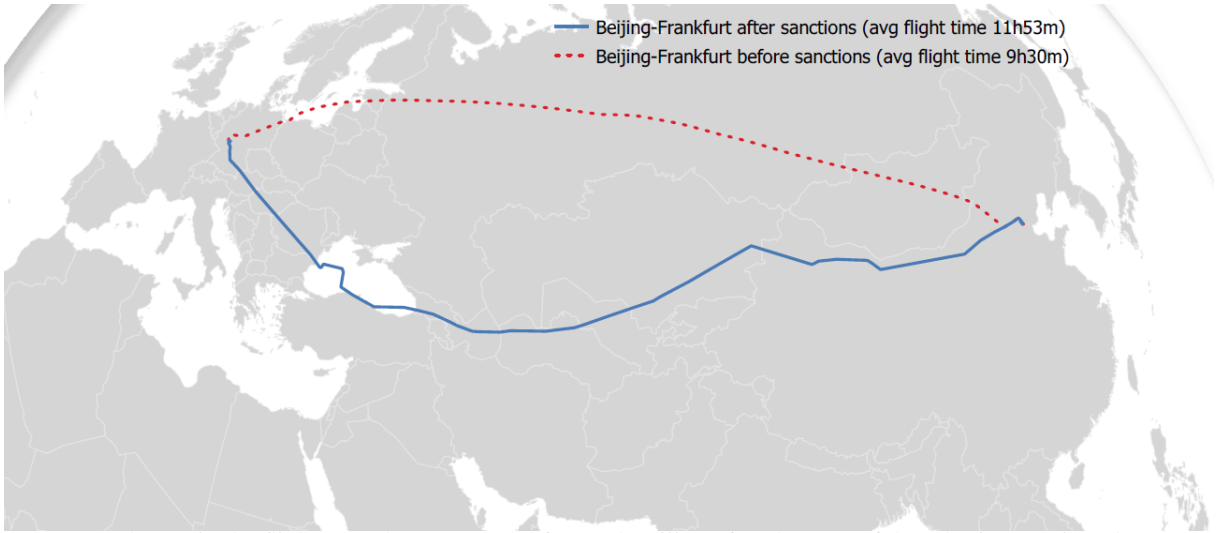
Identification relies on the assumption that, conditional on controls, the airspace closure affects relative trade shares across transport modes only through its impact on relative transport costs. Several features of the setting support this exclusion restriction. First, although geopolitical tensions had been rising since late 2021, the precise timing and scale of the invasion in February 2022 were widely viewed as unexpected, limiting firms' ability to adjust shipping decisions in advance. Second, the disruption proved highly persistent (Zabrodskyi et al., 2022), exhausting inventories and forcing firms to reoptimize logistics rather than rely on short-run inventory smoothing, so the estimated responses reflect medium- to long-run substitution. Third, the shock did not involve direct physical destruction of transport infrastructure along Europe–Asia routes, ruling out supply-side disruptions that could differentially affect

[16] The restrictions also apply to other sanctioning countries, including Japan and South Korea. Chinese carriers, by contrast, continue to operate over Russian airspace.

[17] Affected routes include flights to Japan, South Korea, China, and several destinations in Southeast Asia. Flights are typically diverted southward through Turkey, Central Asia, China, and Mongolia (source: Flightradar24).

[18] Source: Eurostat.

Figure 3: Sanctions and Change in Flight Time



Notes: Change in the flight route between Frankfurt and Beijing after the start of the Ukraine war in February 2022. The flight time increases by 25 % from 9h30mins to 11h53mins. Author's own calculation based on Flightradar24 data.

product availability across modes. Fourth, since rerouted flights do not pass over affected territories, there are no additional security considerations that would differentially favor air or sea transport beyond changes in relative prices.

Finally, sanctions-related trade restrictions other than airspace closures are unlikely to bias the estimates. While sanctions may reduce aggregate trade volumes between some country pairs, they do not differentially affect air and maritime transport costs and therefore should not mechanically shift relative mode shares. The identifying variation thus arises from within country-pair changes in transport mode shares, holding fixed bilateral demand conditions and product composition. This structure supports a causal interpretation of the estimated elasticity as reflecting substitution across transport modes in response to relative transport cost changes.

4.1.1 Empirical Strategy

My identification strategy relies on a difference-in-differences design that compares trade flows shipped by air and sea along routes affected by the war with those that were not, before and after the outbreak of the conflict in February 2022. This approach exploits the exogenous variation generated by the unexpected closure of Russian and Ukrainian airspace. Affected routes face higher air transport costs due to longer flight paths, which increase travel time and operating expenses.[19] Figure 3 illustrates this mechanism. A flight from Frankfurt to Beijing operated by a European carrier, which previously crossed Russian airspace, must now take a longer detour,

[19]For example, a report from the Federal Aviation Administration indicates an average hourly operating cost of about USD 10,000 for a wide-body aircraft with more than 300 seats, over 90% of which consists of variable costs such as fuel and personnel. Fuel alone accounts for nearly 20% of total flight expenditures (FAA, 2022).

increasing flight time by more than two hours.

I use data on daily flights from January 2019 to December 2022, covering more than 1.8 million observations, from Strohmeier et al. (2021). These data are collected from ADS-B signals via the OpenSky platform.[20] The key variables are the origin and destination airports and total flight time. I restrict the sample to international flights, excluding observations in which takeoff and landing occur within the same country, and thus focus on direct flights only. An underlying assumption is that commercial and cargo flights follow similar routes and have comparable travel times.[21] I then compute monthly average flight times between country pairs, which I use as a proxy for air transport costs.

I define a route as an origin–destination country pair and classify it as affected if pre-war flight paths crossed either Russian or Ukrainian airspace. The treated group consists of routes between European countries and destinations in East and Southeast Asia. To identify affected routes, I rely on reports from the International Civil Aviation Organization (ICAO) and supplement them with manual verification using pre-war flight paths from Flightradar24.[22] The control group consists of North and South American countries that trade with Europe.[23]

Finally, I restrict the sample to HS6 sectors for which at least 5% of trade value was shipped by both air and sea in 2019. This restriction allows me to focus on the intensive margin of adjustment rather than the extensive margin. In particular, I exploit variation within origin–destination–sector triplets that already use both modes, excluding cases where trade switches from a single mode to multiple modes. In Appendix D.3, I estimate a linear probability model to verify that the shock did not significantly affect the extensive margin by eliminating air transport for specific origin–destination–sector combinations.

To estimate the elasticity of substitution between air and sea transport following the increase in transport costs induced by the closure of Russian and Ukrainian airspace, I adopt a two-step approach. First, I estimate the effect of the sanctions on average flight times between European countries and Asian destinations:

$$\ln \text{FlightTime}_{nit} = \beta_1 \text{Post}_t \times \text{Treated}_{ni} + \mu_{nih} + \mu_{ht} + \epsilon_{nit}, \tag{16}$$

where FlightTime_{nit} denotes the average flight time between countries n and i at time t, Treated_{ni}

[20] All aircraft are required to carry an ADS-B transponder while in flight.

[21] I validate this assumption using Flightradar24, which reports real-time flight paths. Figure H.6 provides an example for flights between Beijing and Frankfurt.

[22] Affected destinations include China and Hong Kong, Indonesia, Japan, South Korea, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam.

[23] Because the affected routes are long-haul, I focus on air and sea transport only, as substitution towards land transport is limited. Moreover, land transport is not a feasible option for trade with the control group.

is an indicator equal to one if the route is affected by the airspace closure, and Post_t equals one from February 2022 onward. The terms μ_{nih} and μ_{ht} denote origin–destination–sector and sector–time fixed effects, respectively. The coefficient of interest is β_1, which captures the change in average flight time induced by the war.[24]

In the second step, I estimate the impact of the sanctions on relative trade flows across transport modes. To isolate substitution across modes, I control for overall trade volumes within each origin–destination–sector triplet by taking the ratio of air to sea trade flows. This ratio nets out changes in total bilateral trade driven by common shocks and allows me to focus on changes in the relative modal composition of trade. Accordingly, the analysis does not aim to capture the overall effect of the war on trade volumes, but rather its effect on the allocation of trade between air and sea transport. In Appendix D.2, I separately examine the effect of the sanctions on trade flows by mode at the country-pair level.

I estimate the following equation:

$$\ln\left(\frac{X^{Sea}_{niht}}{X^{Air}_{niht}}\right) = \beta_2 \text{Post}_t \times \text{Treated}_{ni} + \mu_{nih} + \mu_{ht} + \epsilon_{niht}, \tag{17}$$

where $\ln\left(X^{sea}_{niht} \,/\, X^{air}_{niht}\right)$ is the relative share of sea trade with respect to air trade between countries n and i in sector h at time t, Treated_{ni}, Post_t, μ_{nih}, and μ_{ht} are defined as above. The coefficient of interest is β_2, which captures the change in sea trade flows with respect to air trade due to the increase in transport costs that followed the closure of the Russian airspace.

Finally, I calculate the ratio of these two coefficients to recover the elasticity of substitution between air and sea transport. This approach is referred to as the Wald-DiD estimator (De Chaisemartin and d'Haultfoeuille, 2023):

$$\beta_{DiD} = \frac{E[\ln(X^{Sea}_{niht}/X^{Air}_{niht})|\text{Post}_t \times \text{Treated}_{ni} = 1] - E[\ln(X^{Sea}_{niht}/X^{Air}_{niht})|\text{Post}_t \times \text{Treated}_{ni} = 0]}{E[\ln \text{FlightTime}_{niht}|\text{Post}_t \times \text{Treated}_{ni} = 1] - E[\ln \text{FlightTime}_{nit}|\text{Post}_t \times \text{Treated}_{ni} = 0]}. \tag{18}$$

The intuition behind this method is that, conditional on a set of controls, changes in average flight time between two countries provide a valid proxy for changes in air transport costs. These changes in flight time are therefore used as an instrument for air transport costs. The identifying assumption is that, once other factors are controlled for, the shock affects the relative trade shares between affected countries only through its impact on flight time.

This approach is equivalent to a two-stage least squares (2SLS) regression of the outcome $\ln\left(X^{Sea}_{niht}/X^{Air}_{niht}\right)$ on the endogenous regressor $\ln \text{FlightTime}_{niht}$, using time variation

[24] Because the sanctions do not affect sea travel times, this coefficient can be interpreted as the relative change in air transport costs with respect to sea transport.

(Post_t), group variation (Treated_{ni}), and their interaction as instruments (De Chaisemartin and d'Haultfoeuille, 2018).

Because this is a sharp difference-in-differences design in which treatment applies only to the treated group, the key identifying assumption is the validity of common trends. This sharp design is justified by the fact that exposure to the airspace closure is determined exogenously by countries' geographic locations.

4.1.2 Identification Assumptions

In this section, I assess whether the results are driven by ex ante differences between treated and control routes or by pre-existing trends.

Before the war, the composition of trade flows by sector and average flight times are similar across affected and unaffected routes, as shown in Table C.5. The unit of observation is the air trade flow or share between origin i and destination n at time t in sector h, where time is measured at the month–year level and sectors are defined at the 6-digit product level. The similarity of moments across the two groups indicates the absence of meaningful ex ante heterogeneity. Figure 5a shows no evidence of differential pre-treatment trends in flight times, with divergence occurring only after the outbreak of the war. Following the invasion, average flight times increase sharply for treated routes, while control routes continue along their pre-war trajectories, supporting a causal interpretation.

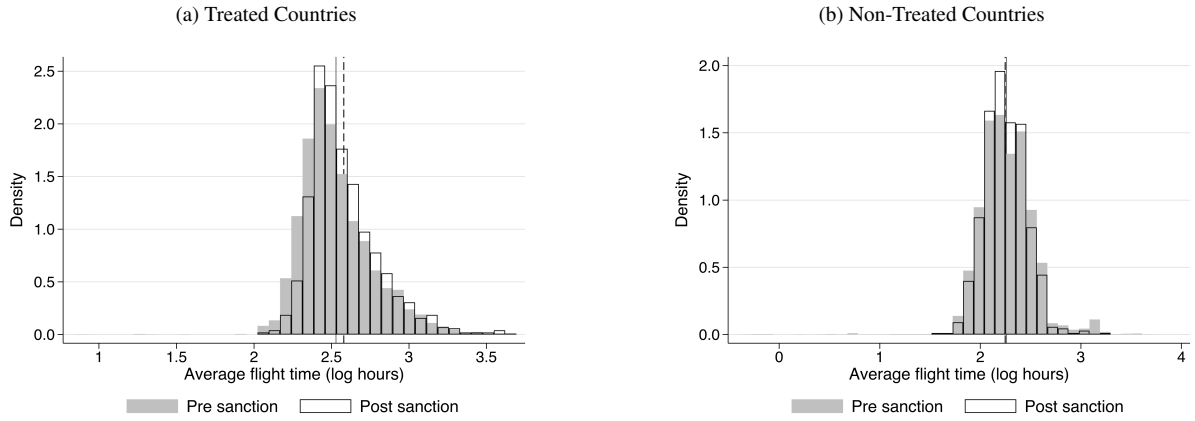
Figure 5b shows that treated routes experience an increase in sea trade relative to air trade after the conflict, again with no evidence of differential pre-treatment trends.[25] An anticipatory response is visible, with air trade beginning to decline in February 2022, prior to the formal onset of hostilities, suggesting that firms responded to rising geopolitical tensions. The robustness checks show that the results are unchanged when February 2022 is classified as part of the pre-treatment period. Finally, Figure 4 confirms that average flight times increase only for treated routes after the invasion, while remaining stable for the control group.

4.1.3 Results

Table 1 reports the estimates of equations (16) and (17). Column (1) shows a significant increase in average flight time on treated routes relative to control routes following the outbreak of the war. The estimated coefficient β_1 is approximately 0.071, implying that average flight time

[25] The same pattern holds when trade is measured in kilograms; see Figure H.7 in the appendix.

Figure 4: Change in the Average Flight Time



Notes: Panel (a) shows the average flight time before and after the sanctions that led to the closure of Russian airspace for treated country pairs. A country pair is classified as treated if it connects a European country with a destination in eastern or southern Asia. Panel (b) presents the corresponding statistics for the control group, which consists of country pairs linking European countries to North and South American destinations. The sample is based on monthly average flight times computed from approximately 1.8 million flight paths observed between January 2019 and December 2022, using AIS data from the OpenSky platform.

rose by about 7% after the onset of the conflict.[26] Column (2) shows that this increase in flight time led to a substantial rise in sea trade relative to air trade. The estimated coefficient of 0.114 indicates that sea trade, measured in euros, increased by about 11% relative to air trade following the outbreak of the war.

The elasticity of substitution between transport modes is obtained using equation (18) by taking the ratio of the coefficient in column (2) to that in column (1). The estimation is implemented using the IV–2SLS strategy described above, and the results are reported in column (3) of Table 1. The results imply that a 1% increase in air transport costs leads to a 1.6% increase in the share of trade shipped by sea. Using equation (8), this corresponds to an implied elasticity of substitution between transport modes of 2.6.

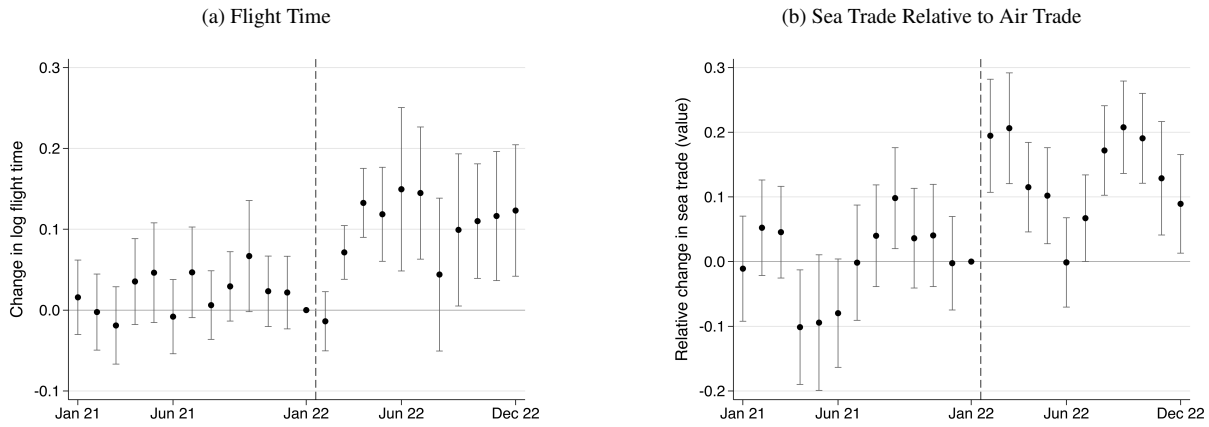
In column (4), I add freight prices for container shipping from Europe to destinations in the Americas and Asia as additional controls.[27] These controls account for potential movements in maritime transport prices over the period. The baseline results remain unchanged.

To the best of my knowledge, this paper is among the first to estimate this elasticity in an international trade context. A related elasticity is estimated by Fuchs and Wong (2024)

[26] Given a pre-war average flight time of roughly 12.5 hours for treated routes, this corresponds to an increase of about one hour. Consistent with this magnitude, Chu et al. (2024) estimates that the closure of Russian airspace raised air transport costs by approximately 12%.

[27] These data are obtained from the Freightos Baltic Index (FBX) via the Eikon platform and consist of monthly averages of daily shipping indices for the following routes: Europe to the West Coast of South America, China to Northern Europe, China to Mediterranean Europe, Europe to the East Coast of North America, Europe to the East Coast of South America, Mediterranean Europe to China, Northern Europe to China, and the East Coast of North America to Europe.

Figure 5: Event Studies



Notes: Panel (a) reports the event-study estimates based on equation (16) and shows no evidence of differential pre-trends in flight time between treated and control country pairs prior to the closure of Russian and Ukrainian airspace. Panel (b) reports the event-study estimates based on equation (17) and shows a significant increase in sea trade relative to air trade following the airspace closures for treated country pairs compared to the control group.

for US domestic trade, where the authors study substitution between rail and truck transport. Their estimates are smaller than those found here, which is consistent with differences in the modes considered. Earlier estimates in Hummels and Schaur (2013) are broadly in line with my findings, though somewhat larger, with maximum values around 6.

In the main analysis, I identify substitution from air to sea transport. In Appendix A, I complement this evidence by exploiting the 2021 disruptions at U.S. West Coast ports as an alternative shock to transport modes, which allows me to estimate substitution from sea to air. Applying the same empirical strategy as in the baseline analysis, I obtain an elasticity of 2.43, which is both qualitatively and quantitatively close to the baseline estimate of 2.6. Taken together, these results support the CES structure adopted in the model, suggesting that a single parameter provides a reasonable characterization of substitution across transport modes.

For comparison, Appendix Table H.17 reports "naive" OLS regressions that yield no significant effects. In that case, identification relies on short-run fluctuations in flight times between country pairs, which are unlikely to induce firms to adjust transport modes. I therefore interpret the baseline estimates as capturing medium-run responses that are sufficiently large to prompt importers to reoptimize their shipping choices in response to persistent trade shocks.

Standard trade models cannot account for these patterns. In conventional gravity frameworks, a single iceberg cost summarizes all bilateral trade frictions. In contrast, the evidence shows that multiple transport modes are used simultaneously along the same origin–destination routes (Figure 2), and their relative shares respond systematically to changes in transport costs. When multiple modes are available, part of the adjustment to higher trade costs occurs through

Table 1: Substitution Between Modes of Transport

	(1) $\ln(\text{FlightTime})$	(2) $\ln(X^{\text{Sea}}/X^{\text{Air}})$	(3) $\ln(X^{\text{Sea}}/X^{\text{Air}})$	(4) $\ln(X^{\text{Sea}}/X^{\text{Air}})$
$\ln(\text{FlightTime})$			1.599	1.891
			(0.697)	(0.745)
Post $\times$ Treated	0.071	0.114		
	(0.015)	(0.031)		
$\ln(\text{Freight})$				0.030
				(0.020)
$\hat{\eta}$			2.60	2.89
Observations	2,821,847	2,821,847	2,821,847	2,821,847
Origin–Destination–HS6 FE	✓	✓	✓	✓
Year–Month–HS6 FE	✓	✓	✓	✓
F-statistic (first stage)	-	-	17.31	18.55
R^2	0.85	0.62	-	-

Notes: Column (1) reports estimates from regression (16), where the dependent variable is the log average flight time for an origin–destination pair. Column (2) reports estimates from regression (17), where the dependent variable is the log ratio (in euros) of sea to air trade between origin country i and destination country n in sector h and month t. Treated equals one for origin–destination pairs involving Asia and zero otherwise. Post equals one after the start of the war (February 2022) and zero otherwise. Columns (3)–(4) estimate the Wald-DiD elasticity in equation (18) via 2SLS; $\hat{\eta}+1$ is the implied elasticity of substitution between air and sea transport. Standard errors are clustered at the origin–destination–year level. Column (4) additionally controls for freight indices from major European ports to Asian and American destinations.

substitution across modes, partially mitigating, but not eliminating, the impact of the shock. As shown in Appendix D.2, air trade declines following the sanctions, while sea trade does not increase sufficiently to offset this reduction, leading to a significant fall in aggregate trade. This pattern is consistent with the predictions of the model.

4.1.4 Product Heterogeneity

Product characteristics can play an important role in shaping the elasticity of substitution between transport modes. I explore this source of heterogeneity in two ways. First, I split the sample according to the ISIC Rev. 3 classification. Second, I partition the sample using several indicators that capture sectoral characteristics of traded goods, including unit value, temperature sensitivity, containerizability (Bernhofen et al., 2016), broad economic classification (BEC), and time sensitivity (Hummels and Schaur, 2013). Table H.13 reports descriptive statistics on the distribution of trade flows, in both value and volume, across these subgroups.

Panel A of Table C.7 presents the 2SLS-IV estimates at the origin–destination–HS6 level, split by ISIC3 sector. As for the pooled estimate, each coefficient equals $\eta_j - 1$ by equation (8), so the implied elasticity is one plus the reported coefficient. These sectoral elasticities are used

as η_j in the quantitative analysis (Section 5.1). Sectors exhibiting relatively high substitutability across transport modes include Textiles (coefficient of 3.58), Manufacturing (3.08), Communications (2.31), and Chemicals (1.83). These sectors tend to comprise goods that are relatively light and high value, making them suitable for shipment by both air and sea. However, splitting the sample into many sectors substantially reduces the number of observations in each subsample, which lowers precision for some estimates.

Panel B of Table C.7 reports results based on product characteristics. Across BEC groups, consumption goods display the highest substitutability, with a coefficient of 2.66. As expected, non-containerizable goods exhibit the lowest coefficient at 0.81. There is no clear pattern regarding temperature sensitivity. In contrast, goods with lower time costs exhibit the highest coefficient at 2.05, which is consistent with the fact that highly time-sensitive goods are rarely shipped by sea.

4.1.5 Robustness Checks

I conduct a series of robustness checks to ensure that the results are not driven by sample selection, timing choices, or unobserved heterogeneity in trade flows. Overall, the findings are robust across alternative samples, timing assumptions, and specifications that allow for within-sector adjustments.

The baseline results are unchanged when trade flows are measured in kilograms rather than euros (Table H.14). Table C.6 shows that the estimates remain robust when using the full sample instead of restricting attention to goods with at least 5% of trade conducted by both modes, and when limiting the sample to manufacturing sectors. To address timing concerns, I redefine the onset of the sanctions as March 2022 rather than February 2022; the estimates remain quantitatively and qualitatively similar. This robustness partly reflects the inclusion of rich fixed effects, which absorb origin–destination–sector triplets with no variation in transport mode use.

I further assess robustness to alternative inference and sample definitions. Re-estimating the baseline specification with standard errors clustered at the origin–destination–HS6 and origin–destination levels yields similar results (Table C.6). Excluding landlocked European countries, such as Austria, the Czech Republic, Hungary, and Slovakia, also leaves the estimates unchanged (Table C.6). To examine potential compositional differences within HS6 sectors, I exploit variation in unit values across modes. Table H.15 shows that substitutability is highest when unit values are similar across modes and becomes insignificant when they differ substan-

tially, consistent with large unit value gaps reflecting fundamentally different products rather than transport choices. Additional tests reported in Table H.16 show that neither unit value ratios nor unit values by mode respond significantly to increases in flight time, ruling out quality upgrading as a driver of the results.

Finally, Appendix D.3 shows that the number of transport modes used changes only marginally following the sanctions, confirming that adjustment occurs primarily along the intensive margin, changes in relative mode shares, rather than through entry or exit of transport modes.

4.2 Congestion

The model introduces a second parameter, λ, which captures the extent to which transport prices respond to changes in trade flows along a given origin–destination–mode. While this parameter is commonly used in spatial models of economic activity within countries (Allen and Arkolakis, 2014, 2022; Fajgelbaum and Schaal, 2020; Fuchs and Wong, 2024), this paper is among the first to incorporate it into a trade model. In this section, I estimate the congestion parameter for air and sea transport using a shift-share instrumental variable approach and show that the resulting estimates are consistent with those found in the literature.

4.2.1 Strategy

In equation (9), transport costs between two countries depend not only on bilateral characteristics, such as distance or mode-specific prices, but also on how intensively a given mode is used along an origin–destination route. This creates a clear endogeneity problem if one were to estimate directly the effect of trade flows on transport prices, since prices and quantities are jointly determined. Identifying λ therefore requires a source of exogenous variation. Unfortunately, because Eurostat does not report transport prices, I cannot apply the strategy used in the previous section to estimate this parameter.

To address this issue, I employ a shift-share instrumental variable design using data from the US Census Bureau on US imports. Specifically, I instrument changes in trade flows to the United States for a given origin–mode–sector (HS6) with changes in trade flows of the same HS6 sector from all other origins exporting to the United States. As an alternative instrument, I also use European exports of the same HS6 sector to non-US destinations. Both shifters capture exogenous demand changes for specific goods along routes that should not directly affect transport prices for a given origin–mode shipping to the United States.

The corresponding shares are constructed as the HS6 share within an HS4 sector for each

origin–mode, measured in 2012, the first year of the sample. This design identifies the congestion parameter λ_m as a share-weighted average of shocks that are plausibly exogenous (Borusyak et al., 2025). Moreover, the approach satisfies the key conditions for shift-share identification outlined by Borusyak et al. (2025): there are many independent shifters, and the shares sum to one.

I estimate the following equation:

$$\ln freight_{omkt} = \beta_\lambda \ln Trade_{omkt} + \mu_{oyk} + \mu_{ot} + \epsilon_{omkt} \tag{19}$$

Where $\ln freight_{omkt}$ is the average freight cost per kilo, computed as total charges divided by total weight, for importing an HS4 product k from origin o to the US using mode m in month t. $\ln Trade_{omkt}$ denotes trade flows, measured in kilos, for the same origin–product combination. The fixed effect μ_{oyk} captures total flows for product k from origin o in year y, while μ_{ot} controls for total flows from country o in month t. Together, these two sets of fixed effects control for total demand in a given year and sector, capturing general increases in US import demand by restricting identification to within-year demand shocks, and for total monthly US demand for goods from country o, which accounts for changes in tariffs or other shocks to aggregate bilateral trade. Standard errors are clustered at the origin level.

The parameter of interest is β_λ, which measures how transport costs respond to increases in trade flows at the origin–product–mode level. Theory predicts this parameter to be positive, as higher trade volumes generate congestion costs. However, equation (19) cannot be estimated directly due to endogeneity, since trade flows and transport costs are jointly determined. I therefore instrument changes in trade flows for a given origin–HS4–mode with a shift-share measure constructed from the sum of imports of HS6 products within the same HS4 category from origins other than o in a given month, weighted by the importance of each HS6 product in total HS4 imports.

$$TradeShift_{omkt} = \sum_{h \in k} \sum_{j \neq o} s_{okm,2012} X_{jhmt} \tag{20}$$

Where $s_{okm,2012}$ denotes the share of HS6 product h within HS4 sector k imported from origin o using mode m in 2012, and X_{jkmt} is US imports of HS6 product h using mode m in month t from an origin $j \neq o$. As an alternative identification strategy, I also construct $TradeShiftEU_{omkt}$, which follows the same shift-share structure but uses exports of EU countries in a given HS6 sector to all destinations other than the United States as the shifter.

In the baseline analysis, I restrict the sample to the period 2014–2019 to ensure sufficient

distance from the base year used to construct the shares and to avoid the COVID-19 period and the subsequent trade disruptions. As a robustness check, I show that the results are not sensitive to this sample restriction.

4.2.2 Results

An increase in trade flows leads to substantial increases in transportation costs for both air and sea transport. Table 2 reports the estimates of equation (19). Column (1) presents results from an OLS specification without an instrumental variable. The estimated coefficient β_λ is negative, contrary to theoretical predictions, which is likely due to endogeneity arising from the joint determination of quantities and prices. Column (2) reports estimates from equation (19) pooling across transport modes, thus capturing the aggregate average effect. Columns (3) and (4) report results separately for air and sea transport. In all three IV specifications, β_λ is positive and statistically significant, indicating that higher trade flows are associated with higher transportation prices. I interpret this positive relationship as reflecting congestion effects and increased demand for transportation services, which can generate delays and raise transport prices. The first-stage results and F-test statistics reported at the bottom of the table support the relevance and strength of the instruments.

The estimates in Table 2 imply an aggregate congestion parameter of 0.079. This value is consistent with existing estimates in the literature, such as Fuchs and Wong (2024), who find a port congestion elasticity of approximately 0.097, and Carballo et al. (2023), who show that processing delays increase transport prices by about 0.06. A key advantage of this approach is that it allows separate identification of congestion effects for air and sea transport. I find that air transport responds more strongly to increases in trade flows, with $\beta_\lambda = 0.097$, compared to sea transport, for which $\beta_\lambda = 0.062$. A plausible explanation is that air transport, which typically relies on smaller carriers and tighter capacity constraints, is more sensitive to demand spikes. This finding is important because it implies that, within the model presented in Section 3, trade flows by mode respond differently even to trade cost changes that are uniform across modes, such as tariffs, thereby introducing an additional margin of adjustment that is not captured in models with a single transport mode.

What this analysis does not capture, however, is the joint empirical effect of substitution across modes and congestion. This limitation is driven by data availability, as Eurostat does not report transport prices directly. In Appendix E.4, I show that the bias introduced in the estimation of η is small for the range of λ values estimated in this section and reported in the

literature.

Table 2: Congestion and Transport Costs

	(1) $\ln(\text{freight}_{omkt})$ OLS	(2) $\ln(\text{freight}_{omkt})$ IV	(3) $\ln(\text{freight}_{omkt})$ IV (Air)	(4) $\ln(\text{freight}_{omkt})$ IV (Sea)
$\ln(\text{Weight})$	-0.162	0.079	0.097	0.062
	(0.005)	(0.009)	(0.013)	(0.011)
Observations	637,610	637,610	345,104	291,324
Origin–Year–HS4–Mode FE	✓	✓		
Origin–Year–HS4 FE			✓	✓
Year–Month–Origin FE	✓	✓	✓	✓
First-stage Coefficient	-	0.44	0.39	0.47
First-stage P-Value	-	0.000	0.000	0.000
F-statistic (first stage)	-	1,401.97	871.29	1,273.41
R^2	0.81	-	-	-

Notes: Column (1) reports OLS estimates for equation (19). Columns (2)–(4) report estimates from a shift-share IV strategy. The sample consists of monthly U.S. import data from 2014 to 2019 obtained from the U.S. Census Bureau. Standard errors, reported in parentheses, are clustered at the origin level. Columns (3) and (4) restrict the sample to air and sea shipments, respectively.

4.2.3 Robustness

I perform a series of robustness checks to ensure that the results are not driven by the choice of instrument, the fixed effects, or the sample definition. Across all alternative specifications, the estimates remain quantitatively similar.

Panel A of table C.8 reports results using a shift-share instrument constructed from changes in exports from European countries to non-US destinations. Both the aggregate estimate and the mode-specific estimates are consistent with the baseline results. In this specification as well, air transport costs are more sensitive to changes in quantities than sea transport costs.

Panel B of table C.8 presents results under alternative fixed-effect structures. Columns (1) to (3) re-estimate the baseline specification using origin–product–quarter fixed effects instead of origin–product–year fixed effects. This specification sharpens identification by focusing on unexpected within-quarter increases in trade flows that may raise transport prices. Columns (4) to (6) instead include product–time fixed effects, μ_{kt}, which control for time-varying, product-specific shocks. Under both sets of fixed effects, the estimates remain close to the baseline in both magnitude and significance.

Finally, Panel C of table C.8 reports results for alternative samples. Columns (1) to (3) extend the sample to include the COVID-19 period and the subsequent recovery, while columns

(4) to (6) include the years 2012 and 2013. Results are close to the baseline. The only notable difference being slightly lower estimated values of λ_m in the latter specification.

5 Quantitative Experiments

I consider two counterfactual scenarios: the closure of Russian–Ukrainian airspace and the closure of the Suez Canal.[28] These scenarios are well suited to illustrating the role of substitution, as each primarily affects a single transport mode and highlights how geopolitical shocks can disrupt international trade through the transport sector.

For each experiment, I compare outcomes from the full model with endogenous transport costs, described in Section 3, to those from two restricted models. The first rules out substitution across transport modes by imposing Cobb–Douglas expenditure shares over transport modes.[29] The second has no transport modes at all, as in a standard multi-sector trade model: every sector is shipped by a single mode, and the mode-specific shock raises the cost of the entire bilateral flow, regardless of how that flow is split across modes in the data. Comparing the model without modes with the model without substitution isolates the role of the observed modal composition of trade, which determines how exposed each country pair is to a mode-specific shock. Comparing the model without substitution with the full model isolates the role of substitution across modes.

5.1 Taking the Model To the Data

The model is calibrated using data for the year 2019. The production and demand structure follows Caliendo and Parro (2015) without input-output linkages across sectors: final demand is Cobb–Douglas with country-specific sector shares α_n^j, each tradable sector combines labor with its own sectoral composite with value added share β_n^j (equations (35) and (36)), a nontradable sector receives the share $1-\alpha_n$ of absorption, and aggregate and sectoral deficits are held fixed as in Dekle et al. (2007). Solving the model in changes requires base year bilateral trade shares π_{ni}^j, modal shares $\pi_{ni}^{j,m}$, the shares α_n^j and β_n^j, the deficits D_n^j, GDP, and the elasticities θ_j, η_j, and λ_m.

Bilateral trade values come from BACI (Gaulier and Zignago, 2010), gross output and value added from UNIDO INDSTAT (UNIDO, 2026), GDP from the World Bank, and the split

[28] The first experiment is motivated by the war in Ukraine, as discussed in the empirical section, and abstracts from other contemporaneous sanctions. The Suez Canal experiment is inspired by recent geopolitical instability in the Red Sea, which has increased travel distances and transport costs.

[29] See Section E.5 for details.

of each bilateral flow between transport modes from the UNCTAD transport costs database (Hoffmeister et al., 2022), which reports free on board trade values by mode of transport. All sources are mapped to the 16 manufacturing sectors of Caliendo and Parro (2015), defined at the ISIC Rev. 3 two digit level. The sample consists of 47 economies, a European bloc (the EU27, Iceland, Norway, and Switzerland), a China bloc (China, Hong Kong, and Macao), referred to as China, and a rest of the world aggregate constructed as a residual, so that $N = 50$. The entities other than the rest of the world account for 94.8% of world manufacturing exports and 93.1% of world GDP. As in the empirical analysis, I focus on air and sea, which carry 68% of the value of trade between entities in the UNCTAD data; modal shares are computed over these two modes, and air accounts for 34% of trade value. Domestic sales, including trade within each bloc, are shipped by a mode whose cost is unaffected by the counterfactual shocks.[30] The base year is an exact equilibrium of the model.

Trade elasticities θ_j are taken from Caliendo and Parro (2015). Mode substitution elasticities are sector specific, where $\hat{b}_j$ is the sectoral estimate reported in Panel A of Table C.7. The set of multimodal sectors $\mathcal{J}_M$ contains the ten sectors whose estimate is positive and statistically significant at the 10% level, with η_j ranging from 1.58 (Metal Products) to 5.13 (Auto). The remaining six sectors (Food, Office, Other Transport, Paper, Wood, and Petroleum) form $\mathcal{J}_S$ and are assumed to be shipped by sea. The congestion parameters are $\lambda_{\text{air}} = 0.097$ and $\lambda_{\text{sea}} = 0.06$, from the estimates in Table 2. Appendix G describes the construction of every object, Table G.11 summarizes the sources, and Table G.12 reports the sector parameters.

5.2 Closure of Russian–Ukrainian Airspace

The war in Ukraine led to the closure of Russian and Ukrainian airspace for European carriers.[31] I model this event as a permanent 12% increase in air transport costs on routes between Europe and East and Southeast Asia, as indicated by Chu et al. (2024).

In the baseline model, welfare falls by 0.067% on average across the ten entities whose air routes are shocked.[32] Such magnitudes are typical in quantitative trade models (Caliendo and Parro, 2015, 2022). Panel A of Table 3 shows that the effect is heterogeneous across countries. The final effect depends on both sectoral composition, and trade shares by mode, of the affect countries. The loss is 6.4% (around 0.01 pp) larger in the model with Cobb–Douglas shares. This indicate that substitution helps in mitigating the negative effect of the shock.

[30] Within blocs and within countries, manufacturing trade is mostly carried by road and rail, which the shocks considered here do not affect directly.

[31] See Section 4.1 for institutional details. Other sanctions are abstracted to isolate the transport cost shock.

[32] Table H.20 reports the results for all entities.

Given the various data sources used in the calibration it is challenging to use this model to benchmark the observed change in the empirical section. However, if I feed a 7% increase in air costs, in line with the rise in flight times in column (1) of Table 1, the ratio of sea over air import rises, on average, by 11.7% which is close to the coefficient reported in Column (2) of Table 1. This indicates that the model boradly reproduces the patterns observed for this shock.

5.3 Permanent Closure of the Suez Canal

The Suez Canal handles roughly 20% of global seaborne trade (Unctad, 2024). Motivated by recent disruptions in the Red Sea, I simulate a permanent closure by increasing sea transport costs along affected routes using the changes in sea distance computed by Feyrer (2021) (Appendix G.9).

In the baseline model, welfare falls by aroun 1% on average across the 50 entities.[33] The average loss is 3.4% larger when modal shares are fixed and 24.2% larger without transport modes. The largest losses occur in the Gulf (Panel B of Table 3): Qatar, the United Arab Emirates, and Kuwait lose between 2.9 and 3.4%, against 0.450% for the European bloc, 0.347% for China, and 0.076% for the United States. Averaged by region, losses are largest in Asia, which includes the Gulf, at 1.456%, and smallest in the Americas, at 0.122%, under all three models.

In both experiments, the losses of an entity reflect the interaction of two characteristics of its trade on the affected routes before the shock. The first is the sectoral composition of that trade: sectors differ in their trade elasticities θ_j, in their mode substitution elasticities η_j, and in whether they are multimodal, which determines both their exposure to a shock to one mode and their ability to shift to another. The second is the share of that trade carried by the shocked mode. Because the two characteristics interact at the level of routes and sectors, countries with similar overall reliance on air transport can experience very different losses.

The comparison of the three models delivers two results. First, substitution attenuates welfare losses in both experiments. Second, the gap between the models without modes and without substitution exceeds the gap between the latter and the baseline, by a factor of about six in the Suez experiment: the modal composition of trade, which determines each country pair's exposure to a shock to one mode, is the main margin, and substitution further reduces losses. In the airspace experiment the factor is about forty, but this mostly reflects the small share of air in trade on the shocked routes, which the model without modes ignores. A single mode model

[33] Table H.21 reports the results for all entities.

therefore overstates the cost of disruptions to one mode, and the overstatement is largest when that mode carries a small share of the affected trade.

Table 3: Welfare Changes for the Most Affected Entities

Entity	Baseline	No Substitution	No Modes
Panel A. Russian–Ukrainian Airspace Closure			
Singapore	−0.157	−0.165 (+5.02)	−0.553 (+251.34)
Malaysia	−0.155	−0.157 (+1.06)	−0.263 (+69.75)
Viet Nam	−0.093	−0.102 (+9.31)	−0.463 (+396.69)
Philippines	−0.054	−0.056 (+5.49)	−0.227 (+323.80)
Myanmar	−0.048	−0.057 (+19.28)	−0.402 (+743.37)
European bloc	−0.042	−0.046 (+9.45)	−0.155 (+268.68)
Korea	−0.039	−0.041 (+6.39)	−0.119 (+209.74)
Japan	−0.027	−0.029 (+5.50)	−0.071 (+160.03)
China	−0.026	−0.029 (+9.34)	−0.123 (+369.17)
Indonesia	−0.025	−0.027 (+6.92)	−0.104 (+317.90)
Panel B. Suez Canal Closure			
Qatar	−3.427	−3.555 (+3.74)	−3.934 (+14.81)
United Arab Emirates	−3.277	−3.474 (+6.02)	−5.265 (+60.67)
Kuwait	−2.876	−2.897 (+0.71)	−3.034 (+5.49)
Mauritius	−2.813	−2.796 (−0.59)	−2.744 (−2.46)
Georgia	−2.454	−2.633 (+7.26)	−3.293 (+34.18)
Singapore	−2.442	−2.515 (+2.98)	−3.064 (+25.48)
Oman	−2.120	−2.198 (+3.70)	−2.522 (+18.98)
Myanmar	−2.024	−2.116 (+4.58)	−2.285 (+12.93)
Saudi Arabia	−1.918	−2.027 (+5.67)	−2.283 (+19.03)
Viet Nam	−1.861	−1.925 (+3.44)	−2.274 (+22.20)
European bloc	−0.450	−0.479 (+6.37)	−0.625 (+38.79)
China	−0.347	−0.359 (+3.50)	−0.427 (+23.00)
United States	−0.076	−0.078 (+2.47)	−0.109 (+43.45)

Notes: Welfare changes in percent relative to the initial equilibrium. In parentheses, the percent difference between the welfare change of the restricted model and that of the baseline; positive values indicate a larger loss. Panel A reports the ten entities whose air routes are shocked, ranked by the baseline welfare change. Panel B reports the ten entities with the largest baseline losses, ranked by the baseline welfare change, followed, below the rule, by EUR, China, and the United States when they are not among the ten (none of the three is among the ten). *Baseline* is the full model of Section 3; *No Substitution* fixes modal shares through Cobb–Douglas aggregation across modes (Section E.5); *No Modes* treats every sector as single mode, so the shock raises the cost of the entire bilateral flow. In Panel A, the shock is a 12% increase in air transport costs on routes between EUR and the other nine entities; in Panel B, it raises sea transport costs in proportion to the increase in sea distance from Feyrer (2021). Parameters: η_j as in Table G.12, $\lambda_{\text{air}} = 0.097$, $\lambda_{\text{sea}} = 0.06$ (Appendix G.9).

6 Conclusion

This paper studies how substitution across transport modes shapes the response of international trade to mode-specific transport cost shocks. Motivated by recent disruptions to air and maritime transport, I develop a quantitative multicountry trade framework that allows agents to choose among multiple transport modes, subject to endogenous congestion costs. The model nests standard Ricardian gravity structures while introducing a new margin of adjustment, mode substitution, that is absent from much of the existing trade literature. Solved in changes, the model requires only one additional set of data relative to a standard Ricardian model, namely initial trade shares by mode for each country pair, and these shares determine each pair's exposure to mode-specific shocks.

Empirically, I provide novel estimates of the elasticity of substitution between transport modes using quasi-natural experiments arising from the closure of Russian airspace and disruptions at U.S. West Coast ports. The results indicate substantial substitutability across modes, with meaningful heterogeneity across products. I also estimate congestion elasticities for both maritime and air transport, documenting that congestion plays an important role in shaping equilibrium transport costs.

Counterfactual simulations show that allowing for transport mode substitution attenuates welfare losses from mode-specific increases in transport costs. The modal composition of trade, which determines each country's exposure to a mode-specific shock, matters even more: a model without transport modes substantially overstates the losses and misallocates them across countries.

From a policy perspective, these findings highlight the importance of evaluating trade and infrastructure policies within a framework that explicitly accounts for transport mode substitution. Mode-specific interventions, such as infrastructure investments, canal closures, or sanctions, can have unintended effects once firms adapt their transport choices, and ignoring this adjustment margin may lead to an overestimation of the welfare losses from transport disruptions. More broadly, policies aimed at improving the resilience of global trade are likely to be more effective when designed in a coordinated manner across transport modes, taking into account congestion, substitution, and heterogeneous exposure across countries. A natural direction for future work is to use the framework to evaluate environmental policies that target the emissions of individual transport modes.

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A US West Coast Port Disruption

This section examines the 2021 U.S. West Coast port disruption as an alternative shock to transport modes and uses it to estimate the elasticity of substitution between air and sea transport. The disruption—driven by pandemic-related demand surges, labor shortages, and dislocations in global shipping networks, led to severe port congestion, container shortages, and sharp increases in maritime transport costs (Hu et al., 2021). These developments significantly affected global supply chains and trade flows, prompting firms to seek alternative shipping options.

Some U.S. trade routes were also affected by bilateral sanctions with Russia, particularly flights connecting the United States with Asian destinations such as Japan, China, and South Korea. Because the war in Ukraine coincided with the port disruption, I exclude 2022 from the sample to avoid confounding shocks that simultaneously affected both air and sea transport. Although the exogeneity of this episode is less clear than in the airspace closure experiment, a key advantage of this setting is the availability of direct data on transport costs by mode, product, and origin, which allows me to estimate cost responses without relying on proxies.

A.1 Empirical Strategy

I employ an empirical strategy analogous to that used for the Russian–Ukrainian conflict, comparing air and sea transport outcomes along affected and unaffected routes before and after the U.S. West Coast port disruption. Identification exploits the unexpected nature of the disruption during the post-COVID recovery period. The maintained assumption is that affected routes experienced a relative increase in maritime transport costs, while air transport costs were less directly impacted.

To document this cost differential, I use U.S. Census Bureau data reporting total transport charges for each shipment. I compute the average transport cost per kilogram for each origin–sector–time observation by dividing total transport charges by shipment weight. I then compare changes in relative transport costs by mode across treated and control groups.

The treated group consists of Asian countries that predominantly ship goods to the United States through West Coast ports.[34] The control group consists of European countries, which primarily ship goods through East Coast ports that were less affected by congestion (Hu et al., 2021). As in the baseline analysis, I restrict the sample to HS6 sectors in which at least 5% of trade value was shipped by both air and sea in 2019.[35]

[34] In the baseline specification, treated countries include China and Hong Kong, Japan, South Korea, Taiwan, and Vietnam.

[35] Because U.S. Census data report only air and sea shipments, land transport is not considered.

I first estimate the effect of the disruption on relative transport costs:

$$\ln\left(\frac{\text{Cost}_{iht}^{sea}}{\text{Cost}_{iht}^{air}}\right) = \beta_1, \text{Disruption}_t \times \text{Treated}_i + \mu_{ih} + \mu_{th} + \epsilon_{iht}, \tag{21}$$

where $\ln\left(\text{Cost}_{iht}^{sea}/\text{Cost}_{iht}^{air}\right)$ is the log ratio of average transport costs per kilogram for shipments from origin country i to the United States in sector h at time t. Treated_i indicates whether the origin country is affected by West Coast congestion, and μ_{ih} and μ_{th} denote origin–sector and sector–time fixed effects, respectively.[36] Disruption_t is an indicator equal to one from January 2021 onward, following Hu et al. (2021). The coefficient β_1 captures the relative increase in maritime transport costs induced by the disruption.

I then estimate the effect of the disruption on relative trade flows by mode:

$$\ln\left(\frac{X_{iht}^{sea}}{X_{iht}^{air}}\right) = \beta_2, \text{Disruption}_t \times \text{Treated}_i + \mu_{ih} + \mu_{th} + \epsilon_{iht}, \tag{22}$$

where $\ln\left(X_{iht}^{sea}/X_{iht}^{air}\right)$ is the log ratio of trade values shipped by sea relative to air. The coefficient β_2 captures the change in relative trade flows in response to the disruption. As in the baseline analysis, I compute the elasticity of substitution between air and sea transport using a Wald difference-in-differences estimator, equivalent to a two-stage least squares (2SLS) approach.

A.2 Identification Assumptions

This subsection evaluates the identifying assumptions underlying the analysis of the U.S. West Coast port disruption, following the same validation steps used for the Russian airspace closure.

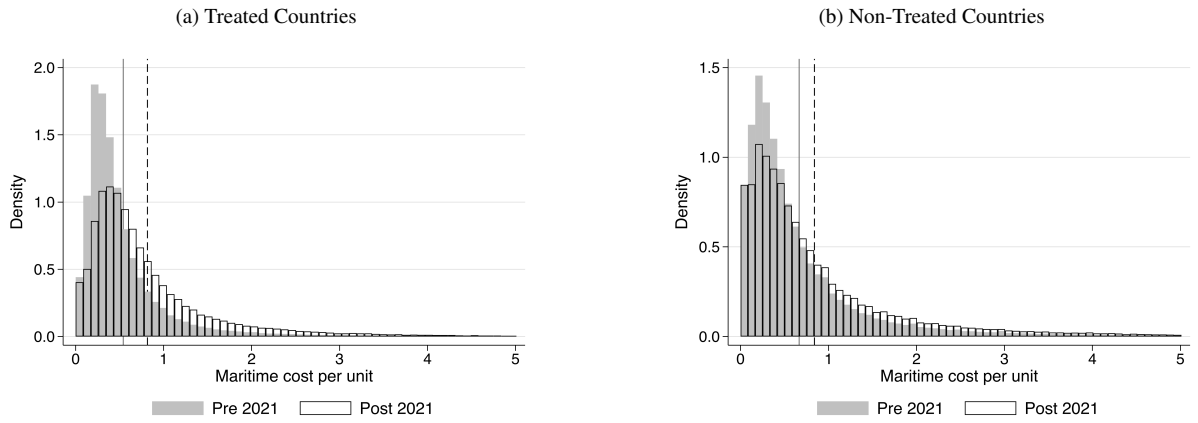
First, Table H.18 shows that treated and control groups were similar in terms of trade values and transport costs prior to the shock. I use data from 2019, which predates both the port disruption and the COVID-19 pandemic. The similarity across groups reduces concerns that differential pre-shock trends or sample composition drive the results.

Second, Figure A.6 illustrates that relative maritime transport costs increased disproportionately for treated countries. The left panel shows a rightward shift in the distribution of sea transport costs for treated origins, indicating higher per-unit shipping costs from Asia. By contrast, the right panel shows relatively stable cost distributions for the control group.

Finally, Figure A.7 presents event-study estimates based on equations (21) and (22). The left panel plots the evolution of relative transport costs, while the right panel plots relative

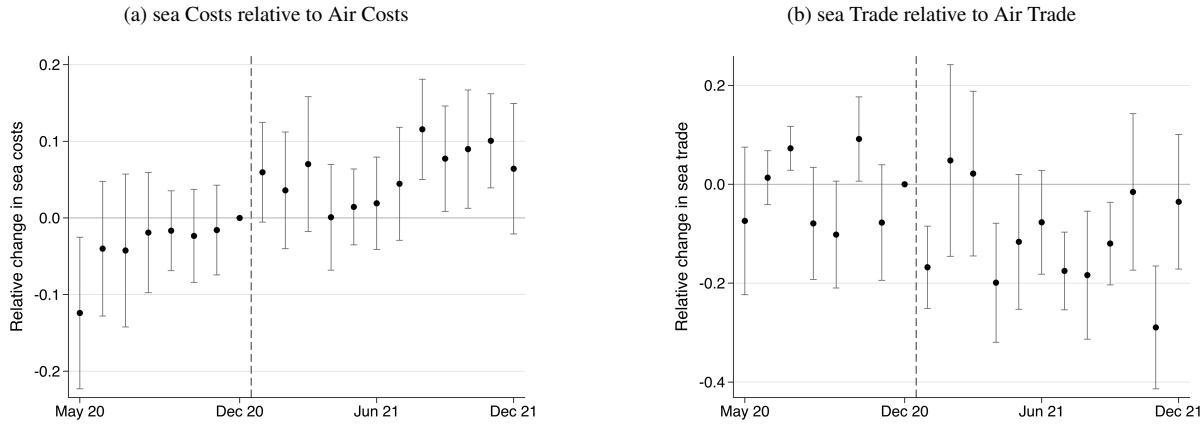
[36] Since the Census data report only U.S. imports, the destination country is always the United States.

Figure A.6: US Data: Change in sea Transport Costs



Notes: Panel (a) shows the distribution of sea costs in the treated group before and after the shock. Panel (b) shows the same for the control group. The treated group experienced a significant increase in sea costs compared to the control group.

Figure A.7: US Data: Event Studies



Notes: Panel (a) reports the event study corresponding to equation (21), while panel (b) reports the event study corresponding to equation (22). The treated group is composed of US trading partners in Asia, while the control group is composed of European partners. We can see that after 2021 when congestion starts to engulf US West Coast ports there is a significant increase in the sea relative price in affected routes. Moreover, the relative trade flows by sea decrease significantly.

trade flows by mode. Both panels indicate that congestion at U.S. West Coast ports led to an increase in relative maritime transport costs and a corresponding decline in sea trade for treated countries. Although the cost series exhibits greater volatility, likely reflecting noise associated with the post-COVID recovery, the absence of differential pre-trends and the alignment of cost and quantity responses support a causal interpretation of the estimates.

A.3 Results

Table A.4 reports the estimation results for equations (21) and (22). Column (1) shows a significant increase in the average transport cost per kilogram for affected countries relative to non-affected countries following the onset of the disruption. The estimated coefficient is 0.121, implying that average maritime transport costs increased by approximately 12% after the dis-

Table A.4: Increase in Sea Transport

	(1) $\ln\left(Freight^{Sea}/Freight^{Air}\right)$	(2) $\ln\left(X^{Sea}/X^{Air}\right)$	(3) $\ln\left(X^{Sea}/X^{Air}\right)$	(4) $\ln\left(X^{Sea}/X^{Air}\right)$	(5) $\ln\left(X^{Sea}/X^{Air}\right)$
ln Freight Ratio		-0.148		-1.432	-1.170
		(0.002)		(0.479)	(0.494)
Post × Treated	0.121		-0.174		
	(0.029)		(0.033)		
$\hat{\eta}$	.	.	.	2.43	2.17
Obs.	1,385,983	1,385,983	1,385,983	1,385,983	1,354,978
Origin-Destination-HS6	✓	✓	✓	✓	✓
Year-Month-HS6	✓	✓	✓	✓	✓
F-Test	.	.	.	17.61	17.28
R^2	0.37	0.70	0.69	-1.17	-0.12

Notes: Column (1) reports estimates from regression (21), where the dependent variable is the log ratio of average sea to air transport costs per kilogram for shipments from origin country i to the United States in sector h and month t. Column (2) reports a "naive" OLS specification in which the log ratio of sea to air trade values is regressed directly on the log freight cost ratio. Column (3) reports estimates from regression (22), where the dependent variable is the log ratio of sea to air trade values. Treated equals one for Asian origins that predominantly ship through U.S. West Coast ports and zero for European origins. Post equals one from January 2021 onward and zero otherwise. Column (4) estimates the Wald-DiD elasticity via 2SLS, which corresponds to the ratio of the coefficients in columns (3) and (1); $\hat{\eta} + 1$ is the implied elasticity of substitution between air and sea transport. Column (5) replicates column (4) measuring trade flows in kilograms rather than values. The sample consists of monthly U.S. import data obtained from the U.S. Census Bureau. Standard errors, reported in parentheses, are clustered at the origin–year level.

ruption.

Column (2) reports a "naive" OLS specification that directly relates monthly changes in relative freight costs to changes in relative trade shares by mode. The estimated coefficient is substantially smaller, suggesting that short-run monthly fluctuations in transport costs generate limited reallocation across modes. Instead, the larger effects captured by the difference-in-differences strategy reflect longer-run adjustments in shipping behavior in response to persistent cost shocks.

Column (3) shows that relative sea trade declined significantly compared with air trade following the disruption. Column (4) reports the implied elasticity of substitution between sea and air transport obtained using the 2SLS (Wald-DiD) strategy, which corresponds to the ratio of the coefficients in columns (3) and (1). Finally, column (5) demonstrates that the results are robust to measuring trade flows in physical quantities rather than values.

This ratio can be interpreted as the effect of a 1% increase in transport costs, proxied by the average cost per kilogram, on the relative share of trade shipped by sea. The estimates reported in the table imply an elasticity of substitution between air and sea transport of approximately 2.43.

Comparing this estimate with the elasticity obtained in Table 1 shows that the two values are remarkably similar, despite being identified in distinct institutional settings and through differ-

ent shocks. This consistency lends support to the CES assumption underlying the model, which posits a single elasticity parameter governing substitution between transport modes across contexts.

B Theory Appendix

B.1 Scope of the Discrete-Choice and Congestion Derivations in the Two-Type Model

The derivations in Sections E.1 and E.3 apply to different subsets of sectors in the model of Section 3.

Section E.1 (Discrete-Choice Derivation). The nested logit model and its CES demand system for transport apply exclusively to multi-mode sectors $j \in \mathcal{J}_M$, for which $M_j \geq 2$ modes are available. For these sectors, the derivation of Section E.1 delivers the transport-cost index (5) and the modal shares (6) with sector-specific elasticity η_j. For single-mode sectors $j \in \mathcal{J}_S$, the discrete-choice problem is trivial: with $M_j = 1$ mode available, the importer faces no modal choice, so there is no lower CES nest, no η_j, and no modal-share equation. The transport cost is given directly by equation (7).

Section E.3 (Derivation of $\hat{d}^j_{ni}$ with Congestion). The derivation of $\hat{d}^j_{ni}$ in Section E.3 applies to multi-mode sectors $j \in \mathcal{J}_M$, yielding equation (12) with $\eta \to \eta_j$ and $\lambda \to \lambda_{j,m}$. For single-mode sectors $j \in \mathcal{J}_S$, equation (12) is not needed; the change in the transport-cost index follows directly from equation (7) as the $M_j = 1$ special case of Section E.3.

Single-mode derivation. At $M_j = 1$, the single mode plays the role of the shocked mode s in equation (12), giving $\pi^s_{ni} = 1$ and an empty $\sum^{-s}$ term. Equation (12) then collapses to

$$\hat{d}^j_{ni} = \left[\left(\hat{\delta}^j_{ni}\,(\hat{X}^j_{ni})^{\lambda_j}\right)^{1-\eta_j}\right]^{1/(1-\eta_j)} = \hat{\delta}^j_{ni}\,(\hat{X}^j_{ni})^{\lambda_j},$$

which is equation (13), with η_j absent. This establishes equation (13) as the $M_j = 1$ restriction of the general derivation of Section E.3, rather than an independent result. The congestion feedback through $\hat{X}^j_{ni}$ is retained; only the modal-substitution margin disappears.

C Additional Tables

In this section I report additional tables from the main text

Table C.5: Comparison of Treated and Control Groups

	Group	Mean	Median	Std. Dev.
Panel A. Air Trade				
Log weight	Non-treated	8.39	8.56	2.26
	Treated	8.73	9.00	2.54
Log value	Non-treated	12.66	12.83	2.64
	Treated	12.64	12.87	2.77
Panel B. Sea Trade				
Log weight	Non-treated	9.35	9.75	3.25
	Treated	10.49	10.97	3.15
Log value	Non-treated	12.35	12.76	2.99
	Treated	13.21	13.58	1.78

Notes: The table compares treated and control groups to assess whether the estimated effects are driven by pre-shock differences in trade composition. The treated group consists of country pairs formed by European countries and destinations in Eastern or Southern Asia. The control group includes all other country pairs. Data are from Eurostat for 2019. The unit of observation is monthly bilateral trade flows at the country-pair–sector level. Sectors are defined at the HS6 level.

Table C.6: Substitution Between Modes of Transport: Robustness

	(1) Full Sample	(2) Manufacture	(3) March 22	(4) Alt Cluster	(5) Alt Cluster 2	(6) No Landlock
ln FlightTime	1.331	1.530	1.220	1.594	1.594	1.593
	(0.583)	(0.589)	(0.456)	(0.107)	(0.701)	(0.615)
$\hat{\eta}$	2.33	2.53	2.22	2.59	2.59	2.59
Obs.	2,187,907	2,592,594	2,768,672	2,768,672	2,768,672	2,679,633
Origin-Destination-HS6	✓	✓	✓	✓	✓	✓
Year-Month-HS6	✓	✓	✓	✓	✓	✓
F-Test	24.75	23.52	24.88	11,997.55	17.33	21.44

Notes: Column (1) reports estimates using the full sample, Column (2) restricts the sample to manufacturing sectors, Column (3) defines the start of the sanctions period as March 2022. Column (4) reports estimates using standard errors clustered at the origin–destination–HS6 level, Column (5) reports estimates using standard errors clustered at the origin–destination level. Column (6) excludes European countries that are landlocked (Austria, Czech Republic, Hungary, and Slovakia). Standard errors are clustered at the origin-destination-year level.

Table C.7: Elasticity by ISIC Rev. 3 Sector and Product Characteristics

Sample		lnftime	(s.e.)	Obs.	Origin-Destination-HS6	Year-Month-HS6	F-Test
Panel A. ISIC Rev. 3 sectors							
Agriculture		1.242	(1.830)	10,392	✓	✓	34.65
Auto		4.127	(1.198)	13,247	✓	✓	88.10
Basic metals		1.734	(0.793)	61,363	✓	✓	253.80
Chemicals		1.834	(0.474)	182,206	✓	✓	742.94
Communication		2.308	(0.877)	58,484	✓	✓	283.02
Electrical		0.864	(0.348)	245,669	✓	✓	1297.11
Food		1.023	(1.509)	17,227	✓	✓	66.39
Machinery n.e.c.		0.594	(0.213)	458,308	✓	✓	2502.85
Manufacturing n.e.c.		3.082	(0.690)	78,728	✓	✓	302.70
Medical		1.354	(0.423)	175,488	✓	✓	834.35
Metal products		0.575	(0.333)	227,257	✓	✓	1062.81
Minerals		2.201	(0.705)	57,572	✓	✓	243.74
Mining		289.214	(19095.535)	316	✓	✓	0.00
Misc		7.708	(7.933)	717	✓	✓	3.48
Office		0.222	(1.234)	26,295	✓	✓	113.55
Other business		31.833	(75.583)	194	✓	✓	0.74
Other transport		-4.480	(8.545)	5,545	✓	✓	2.98
Paper		-0.287	(0.736)	53,453	✓	✓	238.91
Plastic		1.847	(0.544)	77,367	✓	✓	422.02
Textile		3.584	(0.419)	292,813	✓	✓	770.50
Wood		2.012	(1.823)	9,549	✓	✓	51.71
Panel B. Product characteristics							
BEC	Capital	1.046	(0.638)	526,758	✓	✓	25.95
	Intermediate	1.468	(0.523)	1,633,209	✓	✓	27.75
	Consumption	2.660	(1.172)	608,364	✓	✓	11.06
Containerization	Not suitable	0.810	(0.413)	224,487	✓	✓	39.26
	Slightly suitable	4.058	(1.539)	118,457	✓	✓	9.53
	Suitable	1.600	(0.614)	2,425,728	✓	✓	22.65
Temperature sensitivity	Low	1.480	(0.576)	1,884,728	✓	✓	22.38
	Mid-low	2.048	(0.653)	567,606	✓	✓	24.99
	Mid-high	1.359	(0.808)	247,994	✓	✓	23.40
	High	2.105	(0.966)	68,344	✓	✓	24.50
Time cost	Low	2.050	(1.042)	413,731	✓	✓	13.17
	Mid	1.534	(0.646)	834,322	✓	✓	23.32
	High	1.742	(0.563)	641,868	✓	✓	22.87
	Very high	1.367	(0.503)	878,751	✓	✓	28.74

Notes: Panel A reports estimates of equation (18) by ISIC Rev. 3 sector. Panel B reports OLS estimates of equation (18) by product characteristics. The Broad Economic Classification (BEC) categorizes products by final use. Containerizability is from Bernhofen et al. (2016). Temperature sensitivity ranges from 1 (low) to 4 (high). The time-cost index is a discretized version of Hummels and Schaur (2013). The reported coefficient equals $\eta - 1$ by equation (8). All specifications include origin–destination–HS6 and year–month–HS6 fixed effects. Standard errors are clustered at the origin–destination–HS6 level. Panel B estimates are obtained via 2SLS.

Table C.8: Congestion and Transport Costs: Robustness

	(1) IV	(2) IV (Air)	(3) IV (Sea)	(4) IV	(5) IV (Air)	(6) IV (Sea)
Panel A. EU shift-share instrument						
ln(Weight)	0.089	0.108	0.066			
	(0.009)	(0.013)	(0.011)			
Observations	636,249	344,240	290,827			
Origin–Year–HS4–Mode FE	✓					
Origin–Year–HS4 FE		✓	✓			
Year–Month–Origin FE	✓	✓	✓			
First-stage R^2	0.43	0.40	0.44			
F-statistic (first stage)	1,681.04	1,548.57	809.19			
Panel B. Alternative fixed effects						
ln(Weight)	0.110	0.126	0.098	0.113	0.176	0.098
	(0.010)	(0.016)	(0.013)	(0.010)	(0.025)	(0.015)
Observations	621,569	335,332	285,095	633,421	339,831	284,888
Origin–Quarter–HS4–Mode FE	✓					
Origin–Quarter–HS4 FE		✓	✓			
Origin–Year–HS4–Mode FE				✓		
Origin–Year–HS4 FE					✓	✓
Year–Month–Origin FE	✓	✓	✓	✓	✓	✓
Year–Month–HS4 FE				✓	✓	✓
First-stage R^2	0.44	0.39	0.47	0.43	0.37	0.47
F-statistic (first stage)	1,143.67	797.87	862.69	697.71	218.45	709.22
Panel C. Alternative samples						
ln(Weight)	0.086	0.102	0.079	0.029	0.037	0.022
	(0.008)	(0.010)	(0.011)	(0.006)	(0.010)	(0.007)
Observations	948,925	513,171	434,036	2,613,144	1,229,503	1,381,124
Sample	2014–2022	2014–2022	2014–2022	All years	All years	All years
Origin–Year–HS4–Mode FE	✓			✓		
Origin–Year–HS4 FE		✓	✓		✓	✓
Year–Month–Origin FE	✓	✓	✓	✓	✓	✓
First-stage R^2	0.43	0.38	0.45	0.31	0.26	0.34
F-statistic (first stage)	1,898.86	1,123.57	1,501.16	1,376.20	937.69	1,365.08

Notes: This table reports IV estimates of equation (19) using monthly U.S. import data from the U.S. Census Bureau. Panel A uses a shift-share instrument constructed from EU exports to non-U.S. destinations over the same period. Panel B reports robustness to alternative fixed-effects structures (origin–product–quarter versus specifications including year–month–HS4 fixed effects). In Panel C, Columns (1)-(3) reports the results of the estimation for equation (19) using years 2019-2022. Columns (4) to (6) instead report the estimate including an also 2012 and 2014. The sample used consist of US imports at the monthly frequency from the US Census Bureau. Standard errors in parenthesis are clustered at the origin level. Standard errors (in parentheses) are clustered at the origin level. All dependent variables are $\ln(\text{freight}_{omkt})$.

ONLINE SUPPLEMENTARY APPENDICES

D Data Appendix

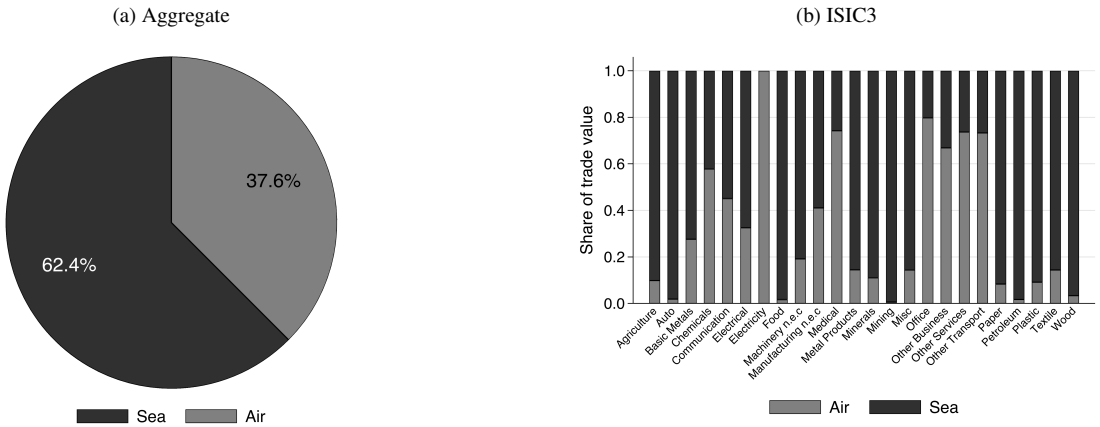
D.1 US Statistics

US Census Bureau data on U.S. imports are widely used in the literature to study transport mode choice in international trade (Hummels, 2007; Hummels and Schaur, 2013) and therefore provide a natural benchmark for assessing the representativeness of the Eurostat data used in the main analysis.

As in European trade, air transport accounts for a substantial share of U.S. imports. Approximately 25% of the total import value is shipped by air (Figure D.8), a pattern that holds across HS sections.

Consistent with the European evidence, Figure D.9 shows that roughly 60% of origin–destination–sector triplets use both air and sea transport. Although this share varies across HS sections, the simultaneous use of both transport modes is observed in all sectors.

Figure D.8: Transport Mode Share of Trade: US Data

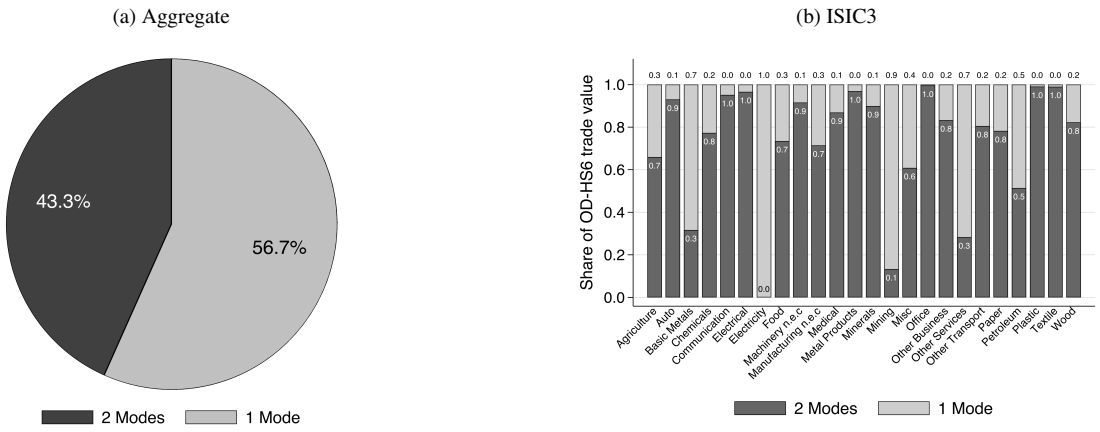


Notes: Panel (a) reports the aggregate share of trade in dollars by mode of transport. Panel (b) reports the aggregate share of trade in dollars by mode of transport by HS Section. The sample consists of US trade flows in euros in 2019, collected from US Census Bureau.

D.2 Aggregate Effect of the Airspace Closure

In this section, I examine the effect of the closure of Russian and Ukrainian airspace on trade flows by transport mode. As shown in the main text, the airspace closure led to a significant increase in average flight times between European countries and Asian destinations. I use this increase in flight time as a proxy for higher transport costs for air-freighted goods. I then

Figure D.9: Number of Modes Used per Origin-HS6: US Data



Notes: Panel (a) reports percentage of origin-product that use two modes of transport. Panel (b) reports percentage of origin-product that use two modes of transport by HS Section. The sample consists of US trade flows in euros in 2019, collected from US Census Bureau.

document that the rise in air transport costs resulted in a decline in the relative share of air transport compared with sea transport. I analyze these effects first at the mode level and then for aggregate trade flows.

To quantify the impact of the airspace closure on trade flows, I estimate the following specification:

$$\ln X_{nihtm} = \beta, \text{Post}_t \times \text{Treated}_{ni} + \mu_{nih} + \mu_{ht} + \epsilon_{nihtm}, \tag{23}$$

where $\ln X_{nihtm}$ denotes the value of trade between exporter i and importer n in sector h at time t, shipped using transport mode m. The variables Treatedni, Post$t$, and the fixed effects μnih and μht are defined as in the main text. I first estimate equation (23) separately for air and sea transport, and then for total trade aggregated across modes.

The coefficient of interest, β, captures the change in trade flows induced by the increase in transport costs following the closure of Russian airspace.

Table D.9 reports the results from estimating equation (23).[37] Column (1) reports estimates for air shipments, while column (2) reports estimates for maritime shipments. The results show that the airspace closure led to a significant decline in trade flows transported by air. The estimated effect on maritime trade is negative but not statistically significant.

Columns (3) and (4) report the effects on aggregate trade flows across transport modes. The aggregate effect is negative and statistically significant, indicating that the increase in transport costs reduced total trade between affected country pairs. This pattern reflects the interaction of two opposing forces. Higher air transport costs generate a positive substitution effect, increasing the relative use of sea transport. At the same time, higher bilateral transport costs give rise

[37]The table reports results for trade flows measured in euros. The results are quantitatively similar when trade flows are measured in kilograms; these estimates are available upon request.

Table D.9: Effects of the Shock on Trade Flows

	(1) Air	(2) Sea	(3) All	(4) Aggregate
Post × Treated	-0.160	-0.038	-0.099	-0.091
	(0.018)	(0.043)	(0.023)	(0.024)
Observations	2,472,709	2,472,709	4,958,592	2,472,709
R^2	0.808	0.772	0.619	0.860
Origin–Destination–HS6 FE	✓	✓	✓	✓
Year–Month–HS6 FE	✓	✓	✓	✓
Transport–HS6 FE			✓	

Notes: This table reports OLS estimates of equation (23). Column (1) uses trade flows shipped by air, while column (2) uses trade flows shipped by sea. Column (3) pools air and sea shipments and includes transport–product fixed effects (μ_{mh}). Column (4) reports results for aggregate trade flows collapsed to the origin–destination–sector level. Standard errors are clustered at the origin–destination level.

to a negative scale effect, lowering overall trade volumes. While substitution toward sea transport raises its share in total trade, it is not strong enough to offset the contraction in aggregate trade. As a result, sea shipments do not increase in absolute terms despite their higher relative importance.

D.3 Airspace Closure and Number of Modes Used

In this section, I examine the effect of the closure of Russian airspace on the number of transport modes used by affected country pairs relative to the control group. The objective is to show that the primary impact of the sanctions operates through changes in mode shares rather than through the abandonment of air transport altogether.

To study this margin, I estimate the following linear probability model:

$$\text{TwoMode}_{niht} = \beta \text{Post}_t \times \text{Treated}_{ni} + \mu_{nih} + \mu_{ht} + \epsilon_{niht}, \tag{24}$$

where TwoMode_{niht} is an indicator equal to one if exporter i and importer n in sector h at time t use both air and sea transport, and zero otherwise. The variables Treated_{ni}, Post_t, and the fixed effects μ_{nih} and μ_{ht} are defined as in the main text. The coefficient of interest, β, captures the effect of the airspace closure on the probability that a given origin–destination–sector triplet uses both transport modes.

Table D.10 reports the estimation results. The closure of Russian and Ukrainian airspace had no economically meaningful effect on the number of transport modes used by affected ori-

gin–destination–sector triplets relative to non-affected ones. Although the estimated coefficient is negative and statistically significant, it corresponds to a reduction in probability of only 1.5 percentage points. This magnitude suggests that the relevant margin of adjustment is the intensive margin, changes in the relative shares of air and sea transport, rather than the extensive margin, defined as the number of modes used.

Table D.10: Increase in air transport: Linear Probability Model

	(1)
	TwoMode_{niht}
Post × Treated	-0.015
	(0.006)
Observations	6,869,061
R-squared	0.603
Origin-destination-HS6	✓
Year-Month-HS6	✓

Notes: This table reports the results of the estimation of equation (24). The dependent variable TwoMode_{niht} is an dummy that takes value 1 if country n and i in sector h at time t used two modes and 0 otherwise. Standard errors are clustered at the origin-destination level. All the estimations are performed as OLS. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

E Additional Theoretical Results

E.1 Solution of the Discrete Choice Model

In this section, I derive the main equations presented in Section 3.2.1. Consumption decisions are modeled as a two-stage process. In the first stage, the consumer chooses the origin of the good; in the second stage, conditional on this choice, she determines the quantity consumed.

Consider a transport mode $m \in M$. The conditional direct utility of a consumer in country n who purchases a good from country i using mode m is given by

$$u_{ni}^m = \ln q_{ni}^m, \tag{25}$$

where q_{ni}^m denotes the quantity consumed. The consumer faces a delivered price

$$p_{ni}^m = p_i d_{ni}^m, \tag{26}$$

where p_i is the factory-gate price and d_{ni}^m denotes the iceberg transport cost associated with shipping from country i to country n using mode m.

Maximizing this utility leads to demand q_{ni}^m. I can now write the indirect utility as

$$V(p_{ni}^m) = \ln p_i + \ln d_{ni}^m - \ln y_n, \tag{27}$$

where y_n is the total expenditure. In the first stage of the problem, I assume that the choice of mode m follows the stochastic utility approach:

$$u_{ni}^m = V(p_{ni}^m) + \mu\epsilon_{ni}^m,$$

with $\mu > 0$ and ϵ_{ni}^m that follow an extreme value type I distribution (Gumbel). Since in this step, the country-pair flow is already decided, I will drop the ni notation for the remainder of the subsection.

The probability density of ϵ^m is $f(\epsilon^m) = \exp(-\epsilon^m)\exp(-\exp(-\epsilon^m))$. The cumulative density function is $F(\epsilon^m) = \exp(-\exp(-\epsilon^m))$. The probability of choosing mode m instead of mode k is:

$$\begin{aligned} Pr(u^m > u^k) &= Pr(V^m + \mu\epsilon^m > V^k + \mu\epsilon^k) \\ &= Pr(V^m - V^k + \mu\epsilon^m > \mu\epsilon^k). \end{aligned}$$

While the conditional probability is

$$Pr(u^m > u^k|\epsilon^m) = F(V^m - V^k + \mu\epsilon^m),$$

and the conditional probability of choosing m given the other options is:

$$Pr(u^m > u^k \forall k \neq m|\epsilon^m) = \prod F(V^m - V^k + \mu\epsilon^m).$$

Finally, the unconditional probability of the mode m is given by

$$Pr(u^m > u^k \forall k \neq m) = \int_{-\infty}^{\infty} \prod_{k \neq m} F(V^m - V^k + \mu\epsilon^m) f(\epsilon^m) d\epsilon^m.$$

Now we have to substitute in the definitions of $F(\cdot)$ and $f(\cdot)$ and solve

$$\begin{aligned}
Pr^m &= \int_{-\infty}^{\infty} \prod_{k \neq m} F(V^m - V^k + \mu\epsilon^m) f(\mu\epsilon^m) d\epsilon^m \\
&= \int_{-\infty}^{\infty} \prod_{k \neq m} \exp(-\exp(V^m - V^k + \mu\epsilon^m)) \exp(-\mu\epsilon^m) \exp(-\exp(-\mu\epsilon^m)) d\epsilon^m \\
&= \int_{-\infty}^{\infty} \prod_{k} \exp(-\exp(V^m - V^k + \mu\epsilon^m)) \exp(-\mu\epsilon^m) d\epsilon^m \\
&= \int_{-\infty}^{\infty} \exp(-\exp(-\mu\epsilon^m) \sum_{k} \exp(V^m - V^k)(-\exp(-\mu\epsilon^m) d\epsilon^m.
\end{aligned}$$

Change of variables to $t = -\exp(-\mu\epsilon^m)$ and $(1/\mu)dt = \exp(-\mu\epsilon^m)$.

$$Pr^m = \int_{-\infty}^{0} \exp(t \sum_{k} \exp(V^m - V^k)) \frac{1}{\mu} dt.$$

We can now solve the integral to obtain:

$$Pr^m = \frac{1}{\sum_k \exp\left(-\left(\frac{V^m}{\mu} - \frac{V^k}{\mu}\right)\right)} = \frac{\exp\left(\frac{V^m}{\mu}\right)}{\sum_k \exp\left(\frac{V^k}{\mu}\right)}, \tag{28}$$

I can substitute equation (27) into equation (28) to have that

$$\begin{aligned}
Pr^m &= \frac{\exp\left(V_{ni}^m/\mu\right)}{\sum_k \exp\left(V_{ni}^k/\mu\right)} \\
&= \frac{\exp\left(-\ln p_{ni}^m/\mu\right) \exp\left(-\ln y/\mu\right)}{\sum_k \exp\left(-\ln p_{ni}^k/\mu\right) \exp\left(-\ln y/\mu\right)} \\
&= \frac{(p_{ni}^m)^{-1/\mu}}{\sum_k (p_{ni}^k)^{-1/\mu}} \\
&= \frac{(p_i d_{ni}^m)^{-1/\mu}}{\sum_k (p_i d_{ni}^k)^{-1/\mu}} \\
\pi_{ni}^m &= \frac{(d_{ni}^m)^{-1/\mu}}{\sum_k (d_{ni}^k)^{-1/\mu}}.
\end{aligned} \tag{29}$$

The share of imports by destination n from origin i shipped using mode m depends solely on relative mode-specific transport costs. Because factory-gate prices and other bilateral trade costs are common across transport modes, they cancel out and do not affect the allocation of trade across modes.

Define the elasticity of substitution between transport modes as:

$$\eta \equiv \frac{\mu + 1}{\mu}. \tag{30}$$

Using this definition, equation (29) can be rewritten as:

$$\pi_{ni}^{m} = \frac{(d_{ni}^{m})^{1-\eta}}{\sum_{k}(d_{ni}^{k})^{1-\eta}} = \frac{X_{ni}^{m}}{X_{ni}}, \tag{31}$$

where the assumption of $\mu > 0$ implies $\eta > 1$.

The share in equation (31) can be derived from a CES utility function:

$$u_{ni} = \left(\sum_{m=1}^{M} (q_{ni}^{m})^{(\eta-1)/\eta} \right)^{\eta/(\eta-1)}, \tag{32}$$

that is maximized under the budget constraint: $\sum^{M} q_{ni}^{m} p_{ni}^{m} = \sum^{M} X_{ni}^{m} = X_{ni}$. Where q_{ni}^{m} is the quantity imported by n from i by each mode, and X_{ni} is the total value imported.

E.2 Share by mode with Congestion

The price index from the discrete random choice mode, which corresponds to the transport costs index in the upper nest, is now given by:

$$d_{ni} = \left(\sum_{m=1}^{M} (d_{ni}^{m})^{1-\eta} \right)^{1/(1-\eta)} = \left(\sum_{m=1}^{M} (\delta_{ni}^{m})^{1-\eta} (\Xi_{ni}^{m})^{1-\eta} \right)^{1/(1-\eta)}. \tag{33}$$

Accordingly, the share of mode s between importer i and exporter j is now given by:

$$\pi_{ni}^{m} = \frac{(d_{ni}^{s})^{1-\eta}}{\sum_{m=1}^{M}(d_{ni}^{m})^{1-\eta}} = \frac{(\delta_{ni}^{s})^{1-\eta}(\Xi_{ni}^{s})^{1-\eta}}{\sum_{m=1}^{M}(\delta_{ni}^{m})^{1-\eta}(\Xi_{ni}^{m})^{1-\eta}}. \tag{34}$$

E.3 Derivation of $\hat{d}_{ni}$ with Congestion

To derive equation (12), consider a shock that affects the transport cost of a single mode $s \in M$. Under this shock, we have that:

$$
\begin{aligned}
\hat{d}_{ni} = d'_{ni}/d_{ni} &= \frac{\left(\sum_{m=1}^{M}(d'^{m}_{ni})^{1-\eta}(\Xi'^{m}_{ni})^{1-\eta}\right)^{1/(1-\eta)}}{\left(\sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}\right)^{1/(1-\eta)}} \\
&= \left(\frac{(d'^{s}_{ni})^{1-\eta}(\Xi'^{s}_{ni})^{1-\eta} + \sum^{-s}(d^{m}_{ni})^{1-\eta}(\Xi'^{m}_{ni})^{1-\eta} \pm \sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}}{\sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}}\right)^{1/(1-\eta)} \\
&= \left(1 + \frac{(d'^{s}_{ni})^{1-\eta}(\Xi'^{s}_{ni})^{1-\eta} - (d^{s}_{ni})^{1-\eta}(\Xi^{s}_{ni})^{1-\eta}}{\sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}} + \frac{\sum^{-s}(d^{m}_{ni})^{1-\eta}(\Xi'^{m}_{ni})^{1-\eta} - \sum^{-s}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}}{\sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}}\right)^{1/(1-\eta)} \\
&= \left(1 + \frac{\left[(\hat{d}^{s}_{ni})^{1-\eta}(\hat{\Xi}^{s}_{ni})^{(1-\eta)} - 1\right](d^{s}_{ni})^{1-\eta}(\Xi^{s}_{ni})^{(1-\eta)}}{\sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}} + \frac{\sum^{-s}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{(1-\eta)}\left[(\hat{\Xi}^{m}_{ni})^{(1-\eta)}\right]}{\sum_{m=1}^{M}(d^{m}_{ni})^{1-\eta}(\Xi^{m}_{ni})^{1-\eta}}\right)^{1/(1-\eta)} \\
&= \left[1 + \left((\hat{d}^{s}_{ni}(\hat{X}^{s}_{ni})^{\lambda})^{1-\eta} - 1\right)\pi^{s}_{ni} + \sum^{-s}\pi^{m}_{ni}\left((\hat{X}^{m}_{ni})^{(\lambda)(1-\eta)} - 1\right)\right]^{1/(1-\eta)}
\end{aligned}
$$

In the last step, I use the definition of the mode share in equation (34) together with the fact that $\hat{\Xi}^{s}_{ni} = (\hat{X}^{s}_{ni})^{\lambda}$, which follows from the assumption that transport infrastructure is fixed.

E.4 Interpretation of η with Congestion

Congestion affects agents' ability to substitute across transport modes and therefore potentially alters the interpretation of the elasticity parameter η estimated in Section 4. Within this framework, the equilibrium share of each transport mode is governed by equation (34). As a result, when congestion costs are present, the elasticity of substitution recovered from equation (6) no longer corresponds exactly to the primitive preference parameter.

Importantly, the presence of congestion induces a downward bias in the estimated elasticity of substitution between modes. Intuitively, higher congestion raises the cost of the mode that experiences increased demand, dampening substitution toward that mode and reallocating some traffic back to alternative modes. This endogenous price response attenuates observed changes in mode shares relative to changes in relative costs, leading the reduced-form elasticity to understate the true substitutability between modes.

This bias can be shown analytically by taking the ratio of equation (34) for two arbitrary transport modes m and s and rearranging terms to express mode shares as a function of relative transport costs:

$$
\ln\left(\frac{X^{m}_{ni}}{X^{s}_{ni}}\right) = \frac{1-\eta}{1-\lambda(1-\eta)}\ln\left(\frac{d^{m}_{ni}}{d^{s}_{ni}}\right),
$$

When the congestion parameter λ approaches zero, the expression converges to the standard

CES relationship, and the bias disappears. Since existing estimates in the literature typically find values of λ close to zero (Allen and Arkolakis, 2022; Fuchs and Wong, 2024), the elasticity identified in Section 4 can be interpreted as a close approximation to the true elasticity of substitution, or more conservatively, as a lower bound.

Using the estimated reduced-form coefficient $\hat{\beta}_{\text{DiD}}$ from Section 4 together with the congestion parameter $\lambda = 0.079$ estimated in the main text, the implied elasticity of substitution between transport modes—once congestion is accounted for—is $\eta \approx 2.83$. [38]

E.5 Model with Cobb-Douglas Shares and Congestion

When Cobb–Douglas shares are combined with congestion effects, as described in Section 3.3, the aggregate trade cost index becomes

$$d_{ni} = \prod_{m=1}^{M} (d_{ni}^{m})^{\pi_{ni}^{m}} = \prod_{m=1}^{M} (\delta_{ni}^{m} \Xi_{ni}^{m})^{\pi_{ni}^{m}}$$

where δ_{ni}^{m} denotes the iceberg component of transport costs and Ξ_{ni}^{m} captures congestion.

Since shares are fixed and the shock affects only mode s, the counterfactual change in aggregate trade costs is given by

$$\hat{d}_{ni} = \left(\hat{\delta}_{ni}^{s} \left(\hat{X}_{ni}^{s} \right)^{\lambda} \right)^{\pi_{ni}^{s}} \prod_{m \neq s} \left(\left(\hat{X}_{ni}^{m} \right)^{\lambda} \right)^{\pi_{ni}^{m}}$$

where λ governs the strength of congestion effects. As in the previous case, the remaining equilibrium conditions in changes coincide with those of the baseline model.

F Production, Trade, and Equilibrium

I assume that firms uses intermediates input in production. In a standard Caliendo and Parro (2015), a producer in sector j of country n combines labor with composites drawn from all J sectors, with input–output coefficients $\gamma_n^{kj} \geq 0$ satisfying $\beta_n^j + \sum_{k=1}^{J} \gamma_n^{kj} = 1$ for each n, j, where $\beta_n^j \equiv V_n^j / Y_n^j \in (0, 1]$ is the country-sector value-added (labor) share. This paper imposes

[38] To see this just solve for:

$$\hat{\beta}_{Did} = -\frac{1-\eta}{1-\lambda(1-\eta)} \quad \rightarrow \quad \eta = 1 - \frac{\hat{\beta}_{Did}}{1+\hat{\beta}_{Did}\lambda}$$

a muted input–output structure by setting all off-diagonal coefficients to zero:

$$\gamma_n^{kj} = 0 \quad \text{for } k \neq j, \qquad \text{so } \gamma_n^{jj} = 1 - \beta_n^j. \tag{35}$$

Each sector therefore uses only its own sectoral composite alongside labor as production inputs, a within-sector roundabout structure. Under restriction (35), the unit cost of the input bundle in sector j of country n is

$$c_n^j = w_n^{\beta_n^j} \left(P_n^j\right)^{1-\beta_n^j}, \tag{36}$$

where w_n is the wage, P_n^j is the price index for sector j in country n, and $\beta_n^j \in (0, 1]$ is the country-sector value-added share. The production structure is identical across $\mathcal{J}_M$ and $\mathcal{J}_S$; the sector-type partition affects only the transport-cost component of delivered costs. This is done to keep the focus on the transportation side of the economy but it can easily adapted to integrate input-output linkages.

F.1 Production and Trade

In each sector j, a manufacturer's productivity in country i for good u is drawn from a Fréchet distribution with country-sector scale A_i^j and shape parameter θ_j, which is also the elasticity of bilateral trade flows with respect to trade costs in sector j, independently across countries, sectors, and goods. Under the muted input–output restriction (35), country i produces sector-j goods at unit input-bundle cost $c_i^j = w_i^{\beta_i^j} (P_i^j)^{1-\beta_i^j}$ from equation (36). The delivered cost of a sector-j good to country n is $c_i^j d_{ni}^j$. Because each country sources goods from the cheapest origin, standard extreme-value aggregation yields the sectoral price index

$$P_n^j = \gamma_j \left[\sum_{i=1}^{N} A_i^j \left(c_i^j \, d_{ni}^j \right)^{-\theta_j} \right]^{-1/\theta_j}, \tag{37}$$

where $\gamma_j \equiv \Gamma\big(1 - (1-\sigma)/\theta_j\big)^{1/(1-\sigma)}$ and $\Gamma(\cdot)$ is the gamma function, and the restriction $\theta_j > \sigma - 1$ ensures a finite price index in both types. Country n's expenditure share on sector-j goods sourced from country i is

$$\pi_{ni}^j = \frac{A_i^j \left(c_i^j \, d_{ni}^j \right)^{-\theta_j}}{\sum_{h=1}^{N} A_h^j \left(c_h^j \, d_{nh}^j \right)^{-\theta_j}}. \tag{38}$$

Both equations hold sector by sector across $\mathcal{J}_M \cup \mathcal{J}_S$; the sector type enters only through the construction of d_{ni}^j.

F.2 Market Clearing

Following the national accounting framework of Dekle et al. (2007) generalized to J sectors, GDP is $Y_n = w_n L_n$ and total final absorption satisfies $X_n = w_n L_n + D_n$, where D_n is the exogenous aggregate trade deficit. Gross output of sector j in country n is Y_n^j, and the sectoral trade deficit $D_n^j \equiv X_n^j - Y_n^j$ satisfies $\sum_{j \in \mathcal{J}_M \cup \mathcal{J}_S} D_n^j = D_n$. Under the muted input–output restriction (35), total expenditure on sector-j goods is $X_n^j = \alpha_n^j X_n + (1 - \beta_n^j) Y_n^j$. Goods-market clearing requires

$$Y_n^j = \sum_{i=1}^{N} \pi_{in}^j \, X_i^j,$$

and aggregate labor income satisfies

$$w_n L_n = \sum_{j \in \mathcal{J}_M \cup \mathcal{J}_S} \beta_n^j Y_n^j + (1 - \alpha_n) X_n,$$

where the second term captures labor income from the non-tradable composite, which uses only labor and has Cobb–Douglas expenditure share $1 - \alpha_n$. Solving the expenditure identity for gross output gives $Y_n^j = (\alpha_n^j X_n - D_n^j)/\beta_n^j$; substituting into the labor-market condition and using the share identity (2) with $\sum_j D_n^j = D_n$ confirms $X_n = w_n L_n + D_n$. Imposing goods-market clearing and collecting terms yields the budget constraint[39]

$$w_n L_n + D_n - \sum_{j=1}^{J} \frac{D_n^j}{\beta_n^j} = \sum_{j=1}^{J} \sum_{i=1}^{N} \pi_{ni}^j \left[\frac{\alpha_i^j}{\beta_i^j} (w_i L_i + D_i) - \frac{1 - \beta_i^j}{\beta_i^j} D_i^j \right], \tag{39}$$

where j sums over $\mathcal{J}_M \cup \mathcal{J}_S$.

F.3 General Equilibrium

Equation (1) states that final utility in country n depends on sectoral composites and the non-tradable good. The structure of each composite reflects the sector type: for $j \in \mathcal{J}_M$, goods are sourced from the cheapest origin and transported via the mode nest with sector-specific elasticity η_j, so the lower CES nest over modes is present; for $j \in \mathcal{J}_S$, goods arrive by a single mode and the lower nest is absent. Substituting the transport-cost specifications into the within-sector

[39] See Dekle et al. (2007) for an explicit derivation of the single-sector analog.

CES composite and then into the Cobb–Douglas upper tier gives the full nested utility:

$$U_n = \left(\prod_{j \in \mathcal{J}_M} \left(\int_0^1 \left[\sum_{i=1}^{N} \left(\sum_{m=1}^{M_j} (q_{ni}^{j,m})^{\frac{\eta_j - 1}{\eta_j}} \right)^{\frac{\eta_j}{\eta_j - 1}} \right]^{\frac{\sigma-1}{\sigma}} du \right)^{\frac{\sigma \alpha_n^j}{\sigma-1}} \right) \left(\prod_{j \in \mathcal{J}_S} \left(\int_0^1 \left[\sum_{i=1}^{N} q_{ni}^{j} \right]^{\frac{\sigma-1}{\sigma}} du \right)^{\frac{\sigma \alpha_n^j}{\sigma-1}} \right) (Q_n^{NT})^{1-\alpha_n}, \tag{40}$$

where $q_{ni}^{j,m}$ is the quantity of sector-j goods imported by country n from country i via mode m (for $j \in \mathcal{J}_M$), q_{ni}^{j} is the corresponding quantity for single-mode sectors (for $j \in \mathcal{J}_S$), Q_n^{NT} is the nontradable composite, and the outer products aggregate across sectors with Cobb–Douglas weights α_n^j.

Given primitives $\left\{ A_i^j,\, \theta_j,\, \eta_j\,(j \in \mathcal{J}_M),\, \lambda_m,\, L_n,\, \beta_n^j,\, \alpha_n^j,\, \delta_{ni}^{j,m}\,(j \in \mathcal{J}_M),\, \delta_{ni}^{j,m_j}\,(j \in \mathcal{J}_S),\, I_n,\, D_n,\, D_n^j \right\}$, an equilibrium is a vector of wages $\{w_n\}$, sectoral price indices $\{P_n^j\}$, and transport-cost indices $\{d_{ni}^j\}$ satisfying:

share of expenditure on sector-j goods sourced from country i (both types):

$$\pi_{ni}^{j} = \frac{A_i^j \left(c_i^j \, d_{ni}^j \right)^{-\theta_j}}{\sum_{h=1}^{N} A_h^j \left(c_h^j \, d_{nh}^j \right)^{-\theta_j}},$$

sectoral price index (both types):

$$P_n^j = \gamma_j \left[\sum_{i=1}^{N} A_i^j \left(c_i^j \, d_{ni}^j \right)^{-\theta_j} \right]^{-1/\theta_j},$$

share of mode m in sector-j bilateral trade between n and i (for $j \in \mathcal{J}_M$ only):

$$\pi_{ni}^{j,m} = \frac{\left(d_{ni}^{j,m} \right)^{1-\eta_j}}{\sum_{k=1}^{M_j} \left(d_{ni}^{j,k} \right)^{1-\eta_j}},$$

sector-j aggregate transport-cost index for $j \in \mathcal{J}_M$:

$$d_{ni}^{j} = \left(\sum_{m=1}^{M_j} \left(d_{ni}^{j,m} \right)^{1-\eta_j} \right)^{1/(1-\eta_j)},$$

sector-j transport cost for $j \in \mathcal{J}_S$:

$$d_{ni}^{j} = \frac{\delta_{ni}^{j,m_j} \left(X_{ni}^{j} \right)^{\lambda_{m_j}}}{I_n \, I_i},$$

and the budget constraint with $X_n = w_n L_n + D_n$ and deficits $\{D_n, D_n^j\}$ exogenous:

$$w_n L_n + D_n - \sum_{j=1}^{J} \frac{D_n^j}{\beta_n^j} = \sum_{j=1}^{J} \sum_{i=1}^{N} \pi_{ni}^j \left[\frac{\alpha_i^j}{\beta_i^j} (w_i L_i + D_i) - \frac{1 - \beta_i^j}{\beta_i^j} D_i^j \right].$$

The transport-cost indices $\{d_{ni}^j\}$ are jointly determined with wages and price indices The budget constraint aggregates across $\mathcal{J}_M \cup \mathcal{J}_S$ with country-sector parameters $\{\beta_n^j, \alpha_n^j\}$; cross-sector linkages through intermediate-input costs are absent by the muted input–output restriction (35). The restriction $\theta_j > \sigma - 1$ ensures a finite price index in both types. For $j \in \mathcal{J}_M$, the additional restriction $\eta_j > \sigma$ is required conditional on the source-country choice; no counterpart restriction applies to $j \in \mathcal{J}_S$. There is no restriction between θ_j and η_j since factory-gate prices are independent of the transport mode.

G Details on Calibration

This appendix documents the construction of the base year objects used to solve the multi-sector model of Section 3 in changes. The base year is 2019. All monetary values are in millions of current U.S. dollars. Table G.11 lists the sources, Table G.12 the sector level parameters, and Table H.19 the entities.

G.1 Model Objects and Data Sources

The change form system in Section 3.4 requires, for every importer n, exporter i, sector j, and mode m, the bilateral expenditure shares π_{ni}^j, the modal shares $\pi_{ni}^{j,m}$, the final demand shares α_n^j, the value added shares β_n^j, the sectoral deficits D_n^j, GDP $w_n L_n$, and the elasticities θ_j, η_j, and λ_m. Four data sources deliver these objects. Bilateral trade values are from BACI (Gaulier and Zignago, 2010), HS 2017 nomenclature, version 202601, reported in thousands of U.S. dollars at the HS six digit level. Trade values by mode of transport are from the UNCTAD transport costs database (Hoffmeister et al., 2022), which reports free on board values by origin, destination, HS four digit heading, and mode. Gross output and value added are from UNIDO INDSTAT at the two digit level of ISIC Rev. 3 (UNIDO, 2026). GDP in current U.S. dollars is from the World Bank World Development Indicators. HS codes are mapped to ISIC Rev. 3 with the WITS concordance between the combined HS and ISIC Rev. 3. For UNCTAD, each HS four digit heading is assigned to a single ISIC division.

G.2 Sectors

The tradable sectors are the manufacturing sectors of Caliendo and Parro (2015): Food (ISIC 15 and 16), Textile (17 to 19), Wood (20), Paper (21 and 22), Petroleum (23), Chemicals (24), Plastic (25), Minerals (26), Basic Metals (27), Metal Products (28), Machinery n.e.c. (29), Office (30), Electrical (31), Auto (34), Other Transport (35), and Manufacturing n.e.c. (36 and 37).

G.3 Entities

Countries are grouped into model entities before any screening. The European bloc (EUR) contains the 27 members of the European Union, Iceland, Norway, and Switzerland; the United Kingdom is a separate entity. China (CHN) contains China, Hong Kong, and Macao. A country outside the two blocs enters as a standalone entity if World Bank GDP is available, UNIDO reports output for at least six of the 16 sectors, and the export weighted coverage of UNIDO output,

$$\text{cov}_n = \frac{\sum_j E_n^j \mathbb{1}\{Y_n^j \text{ reported}\}}{\sum_j E_n^j}, \tag{41}$$

is at least 0.75, where E_n^j denotes BACI exports. Twenty one small or landlocked economies are assigned to the rest of the world (ROW) regardless of the screen: Afghanistan, Armenia, Azerbaijan, Bermuda, Bolivia, Botswana, Fiji, Iraq, Jordan, Kazakhstan, the Kyrgyz Republic, Malawi, Mongolia, Montenegro, Nepal, North Macedonia, Paraguay, Rwanda, the State of Palestine, Turkmenistan, and Uzbekistan. All other countries belong to ROW. A standalone entity is moved to ROW if, after the output adjustments of Appendix G.6, its domestic sales are not positive in some sector; this applies to Albania. The final sample has 47 standalone economies, EUR, CHN, and ROW, so $N = 50$ (Table H.19). Measured in BACI, the 47 standalone economies account for 42.0 percent of world exports in the 16 sectors, EUR for 37.9 percent (including trade among its members), CHN for 14.9 percent, and ROW for 5.2 percent. The entities other than ROW produce 93.1 percent of world GDP.

G.4 Bilateral Trade, Domestic Sales, and Rest of the World

For $n \neq i$, bilateral imports of entity n from entity i in sector j sum BACI flows across member countries,

$$X_{ni}^j = \sum_{c \in n} \sum_{c' \in i} \text{BACI}_{c'c}^j, \tag{42}$$

where $\text{BACI}^j_{c'c}$ is the value of exports from country c' to country c. Trade among members of the same entity is not part of X^j_{ni} and enters domestic sales instead. Because ROW collects every BACI country outside the other 49 entities, its trade equals world trade minus the trade of the other entities, and sectoral deficits add up to zero across entities, $\sum_n D^j_n = 0$ for every j. Exports and imports are $E^j_n = \sum_{i \neq n} X^j_{in}$ and $M^j_n = \sum_{i \neq n} X^j_{ni}$. Domestic sales are gross output net of exports,

$$X^j_{nn} = Y^j_n - E^j_n, \tag{43}$$

and the expenditure shares in equation (38) are $\pi^j_{ni} = X^j_{ni} / \sum_h X^j_{nh}$. Of the 39,200 off diagonal entity pairs and sectors, 31,045 have positive trade; the remainder are zeros. Trade between entities amounts to USD 9,109 billion.

G.5 Modal Shares

The modal share of sector j trade from i to n is the ratio of summed free on board values,

$$\pi^{j,\text{air}}_{ni} = \frac{\sum_{c \in n} \sum_{c' \in i} F^{j,\text{air}}_{c'c}}{\sum_{c \in n} \sum_{c' \in i} \left(F^{j,\text{air}}_{c'c} + F^{j,\text{sea}}_{c'c} \right)}, \qquad \pi^{j,\text{sea}}_{ni} = 1 - \pi^{j,\text{air}}_{ni}, \tag{44}$$

where $F^{j,m}_{c'c}$ is the UNCTAD free on board value shipped by mode m. Air and sea carry 68.1 percent of the value of trade between entities in the UNCTAD data; land, pipeline, and other modes are excluded and the shares are computed over air and sea only. The shares are observed for 27,427 entity pairs and sectors, which account for 92.2 percent of trade value. For the remaining cells I impute the trade weighted average share of the same sector across all observed routes between the exporter's and the importer's United Nations subregions, with EUR, CHN, and ROW treated as separate groups (4,324 cells, 7.8 percent of trade value) and, failing that, the trade weighted average of the sector across all observed routes (47 cells, less than 0.01 percent of trade value). Pairs with zero trade receive a share of one half, which does not affect the equilibrium. Domestic sales of multimodal sectors are assigned to a single mode whose cost is unaffected by the shocks. Air accounts for 33.7 percent of the value of trade between entities and for 35.1 percent in the multimodal sectors.

G.6 Gross Output and Value Added Shares

Value added shares. For each entity and sector, $\beta^j_n = \sum_{c \in n} V^j_c / \sum_{c \in n} Y^j_c$, the ratio of summed UNIDO value added to summed gross output. The ratio is used when value added

is positive and $\beta_n^j \in [0.05, 1)$. Otherwise β_n^j is imputed in order: for standalone economies, the median across standalone economies of the same United Nations subregion and sector; for all entities, the median of the sector across entities other than ROW; and, as a last resort, the overall median. After ROW absorbs the entities moved in Appendix G.3, ROW cells without a valid ratio receive the sector median across all entities. Of the 800 entity and sector cells, 731 use the observed ratio (92.6 percent of gross output), 50 the subregion median (2.1 percent), 17 the sector median (3.8 percent), and 2 the post reconstruction sector median (1.5 percent). The final β_n^j range from 0.05 to 0.97, with a median of 0.33.

Gross output. Cells with no UNIDO output are set to 10^{-6}, so that prices remain defined for sectors with no production. For standalone economies that export in such a sector, gross output is imputed as exports times the ratio of output to exports of the same sector in the same subregion, and is never set below exports (38 cells). ROW output is the UNIDO output of the countries in ROW; in four sectors it falls short of ROW exports and is raised to exports times the ratio of output to exports of the same sector among the other entities. The same rule applies to any remaining cell of any entity with $Y_n^j < E_n^j$ (108 cells), which guarantees nonnegative domestic sales in equation (43). A final adjustment, described in Appendix G.7, ensures positive final demand (35 cells). In total 181 cells, which hold 15.9 percent of gross output, are adjusted, and world gross output in the calibration exceeds UNIDO output by 12.5 percent.

GDP. Entity GDP sums World Bank GDP across member countries. ROW GDP is World Bank world GDP minus the GDP of the other 49 entities.

G.7 Deficits, Final Demand Shares, and Base Year Consistency

Sectoral and aggregate deficits, absorption, and sectoral expenditure follow from trade, output, and GDP:

$$D_n^j = M_n^j - E_n^j, \qquad D_n = \sum_j D_n^j, \qquad X_n = w_n L_n + D_n, \qquad X_n^j = Y_n^j + D_n^j. \tag{45}$$

The aggregate deficit D_n is the deficit in the 16 manufacturing sectors; trade in other goods and services is part of the nontradable block. Under the own sector roundabout structure of equation (35), sector j in country n purchases $(1-\beta_n^j)Y_n^j$ of its own composite as intermediate input, so total expenditure on sector j goods is $X_n^j = \alpha_n^j X_n + (1-\beta_n^j)Y_n^j$. Combining this

condition with the definition of the sectoral deficit in equation (4) yields the final demand share

$$\alpha_n^j = \frac{\beta_n^j Y_n^j + D_n^j}{X_n}, \qquad \alpha_n = \sum_j \alpha_n^j. \tag{46}$$

Summing equation (46) over sectors gives $(1-\alpha_n)X_n = w_n L_n - \sum_j \beta_n^j Y_n^j$: the nontradable sector absorbs GDP net of manufacturing value added, which is the labor market condition of Appendix F. Because $\sum_h X_{nh}^j = X_{nn}^j + M_n^j = X_n^j$ by equation (43), goods market clearing $Y_n^j = \sum_i \pi_{in}^j X_i^j$ also holds in the data. The base year is therefore an exact equilibrium: evaluated at $\hat{w}_n = 1$, the residual of the budget constraint (39) is below 10^{-15} of GDP for every entity, so no calibration residual is carried into the counterfactuals.

Equation (46) requires $\beta_n^j Y_n^j + D_n^j > 0$, that is, intermediate use of the own composite cannot exceed domestic absorption of the sector. This condition fails for some large net exporters with low value added shares, mostly in Food, Manufacturing n.e.c., Petroleum, and Textile. In those cells I raise gross output to the smallest value that delivers $\alpha_n^j = 10^{-6}$, $Y_n^j = (10^{-6} X_n - D_n^j)/\beta_n^j$; the adjustment affects 35 cells, which hold 7.2 percent of gross output in the calibration. The share α_n ranges from 0.08 (India) to 0.58 (Singapore), with a median of 0.21. The construction is checked against six conditions: positive domestic sales, $\beta_n^j \in [0.05, 1)$, $\alpha_n^j > 0$, $\alpha_n \in (0,1)$, the identity (46), and $\sum_n D_n^j = 0$; all hold in the final data.

G.8 Parameters

Trade elasticities. The values of θ_j are the estimates of Caliendo and Parro (2015, Table 1, 99 percent sample), with their sector Other mapped to Manufacturing n.e.c. They range from 0.39 (Other Transport) to 64.85 (Petroleum).

Mode substitution elasticities. Panel A of Table C.7 reports, for each sector, the Wald difference in differences coefficient $\hat{b}_j$, the response of $\ln(X^{\text{sea}}/X^{\text{air}})$ to $\ln d^{\text{air}}$. By equation (8), $\hat{b}_j = \eta_j - 1$, so $\eta_j = 1 + \hat{b}_j$, the same mapping that turns the pooled coefficient of 1.6 into $\eta = 2.6$ in Section 4. The multimodal set $\mathcal{J}_M$ contains the ten sectors whose coefficient is positive and statistically significant at the 10 percent level: Auto, Basic Metals, Chemicals, Electrical, Machinery n.e.c., Manufacturing n.e.c., Metal Products, Minerals, Plastic, and Textile. Their elasticities range from 1.58 to 5.13 and all satisfy $\eta_j > 1$.

Congestion. The congestion elasticities are $\lambda_{\text{air}} = 0.097$ and $\lambda_{\text{sea}} = 0.06$ (Table 2). Sectors in $\mathcal{J}_S$ use λ_{sea} in equation (13).

Table G.11: Data Sources and Parameters for Counterfactual Analysis

Object	Source	Classification	Value or unit	Year
Panel A. Trade				
Bilateral trade X_{ni}^{j}	BACI	HS6 2017	USD million	2019
Modal shares $\pi_{ni}^{j,m}$	UNCTAD	HS4	FOB values	2019
Domestic sales X_{nn}^{j}	UNIDO minus BACI	ISIC Rev. 3	USD million	2019
Panel B. Production				
Gross output Y_{n}^{j}, value added	UNIDO INDSTAT	ISIC Rev. 3, two digit	USD million	2019
Value added share β_{n}^{j}	Constructed, Appendix G.6	ISIC Rev. 3	0.05 to 0.97	2019
Panel C. Country Aggregates				
GDP $w_n L_n$	World Bank WDI		Current USD	2019
Deficits D_{n}^{j}, D_n	Constructed from BACI	ISIC Rev. 3	USD million	2019
Final demand share α_{n}^{j}	Constructed, equation (46)	ISIC Rev. 3	α_n: 0.08 to 0.58	2019
Panel E. Parameters				
θ_j	Caliendo and Parro (2015, Table 1)	16 sectors	0.39 to 64.85	
η_j, $j \in \mathcal{J}_M$	Own estimates, Table C.7	10 sectors	1.58 to 5.13	
λ_{air}	Own estimates, Table 2		0.097	
λ_{sea}	Own estimates, Table 2		0.060	

Notes: BACI values are converted from thousands to millions of U.S. dollars and UNCTAD and UNIDO values from U.S. dollars to millions. HS codes are mapped to ISIC Rev. 3 with the WITS concordance. Trade among members of the European bloc (EUR: EU27, Iceland, Norway, Switzerland) and China (CHN: China, Hong Kong, Macao) is treated as domestic. Modal shares are computed over air and sea, which carry 68.1 percent of the value of trade between entities in the UNCTAD data. ROW trade, output, and GDP are residuals with respect to world totals.

G.9 Counterfactual Shocks

Closure of the Suez Canal. For each country pair, the shock to sea transport costs is the ratio of the sea distance with the canal closed to the sea distance with the canal open, from Feyrer (2021): $\hat{\delta}_{c'c}^{\text{sea}} = \text{dist}_{c'c}^{\text{closed}} / \text{dist}_{c'c}^{\text{open}} \geq 1$. Sea transport costs are thus assumed to rise one for one with sea distance. The entity pair shock is the simple average across member country pairs with distance data, and pairs without data receive no shock. Because Egypt borders the canal and can ship from both its Mediterranean and its Red Sea ports, all routes to and from Egypt receive no shock. The shock is common to all sectors and leaves air costs unchanged. It affects 658 of the 2,450 entity pairs, with a mean of 1.163 across all pairs and a maximum of 8.98.

Closure of Russian airspace. Air transport costs increase by 12 percent, $\hat{\delta}_{ni}^{\text{air}} = 1.12$, on routes in both directions between EUR and China, Indonesia, Japan, Korea, Malaysia, Myanmar, the Philippines, Singapore, and Viet Nam (18 entity pairs), in line with the increase in air transport costs documented by Chu et al. (2024). Sea costs are unchanged.

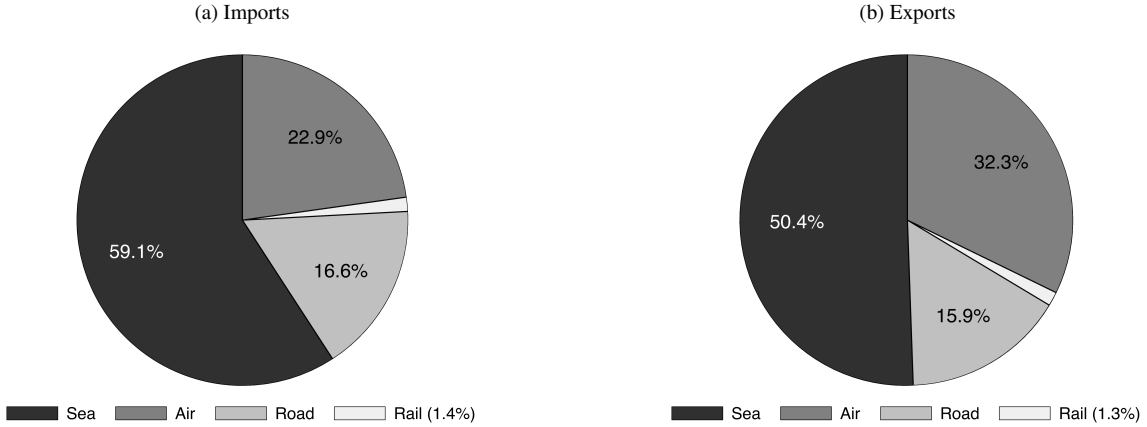
Table G.12: Sector Parameters and Modal Shares

Sector	ISIC Rev. 3	Trade share	θ_j	η_j
Panel A. Multimodal sectors, $\mathcal{J}_M$				
Auto	34	11.1	1.84	5.127
Basic Metals	27	10.3	3.28	2.734
Chemicals	24	14.0	3.13	2.834
Electrical	31	5.8	12.91	1.864
Machinery n.e.c.	29	11.6	1.45	1.594
Manufacturing n.e.c.	36, 37	4.2	3.98	4.082
Metal Products	28	3.1	6.99	1.575
Minerals	26	1.4	2.41	3.201
Plastic	25	3.6	1.67	2.847
Textile	17 to 19	8.6	8.10	4.584
Panel B. Single mode sectors, $\mathcal{J}_S$ (shipped by sea)				
Food	15, 16	7.2	2.62	
Office	30	4.1	12.95	
Other Transport	35	5.6	0.39	
Paper	21, 22	1.8	16.52	
Petroleum	23	7.4	64.85	
Wood	20	0.4	11.50	

Notes: Trade share is the sector's share of the value of trade between entities in the calibration (percent). θ_j is from Caliendo and Parro (2015, Table 1, 99 percent sample). $\eta_j = 1 + \hat{b}_j$, where $\hat{b}_j$ is the sectoral Wald difference in differences estimate in Panel A of Table C.7.

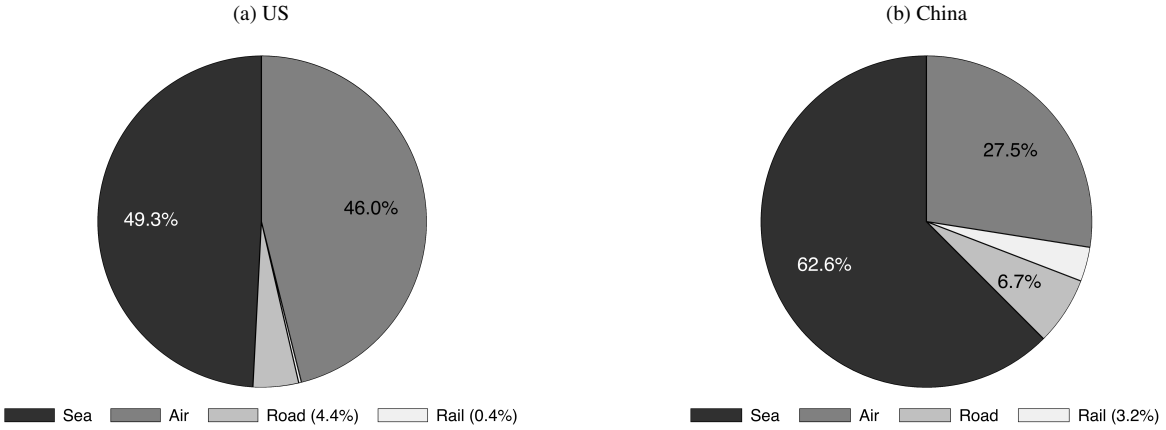
H Additional Figures

Figure H.1: Transport Mode Share of Trade



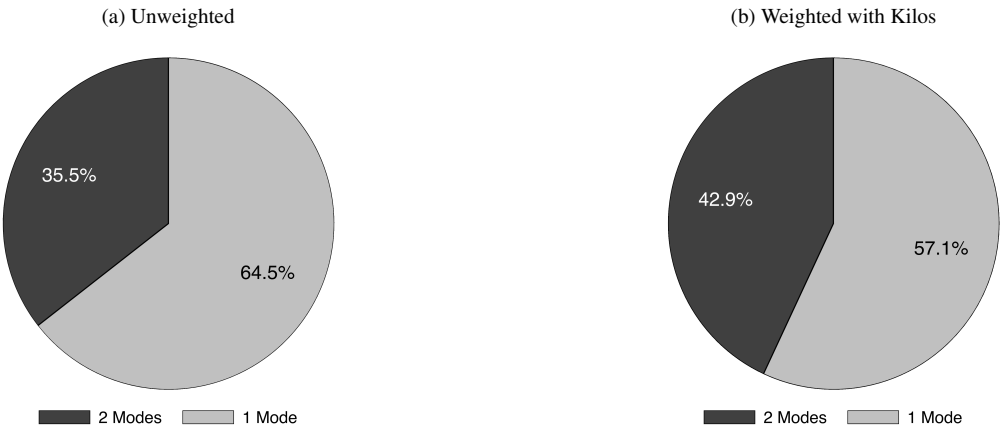
Notes: Panel (a) reports the aggregate share of imported trade in euros by mode of transport. Panel (b) reports the aggregate share of exported trade in euros by mode of transport. The sample consists of European Union members' trade flows in euros in 2019, collected from Eurostat.

Figure H.2: Transport Mode Share of Trade



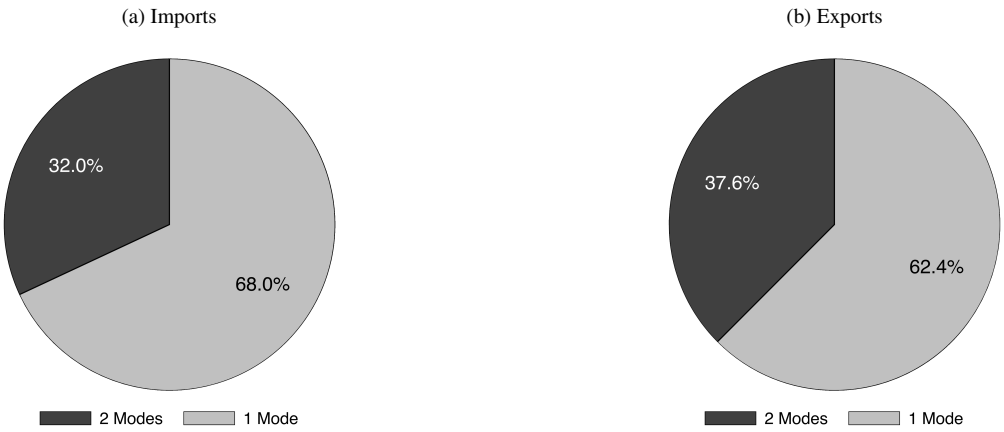
Notes: Panel (a) reports the aggregate share of trade in euros by mode of transport with the United States. Panel (b) reports the aggregate share of trade in euros by mode of transport with China. The sample consists of European Union members' trade flows in euros in 2019, collected from Eurostat.

Figure H.3: Number of Modes Used per Origin-Destination-HS6: Alternative



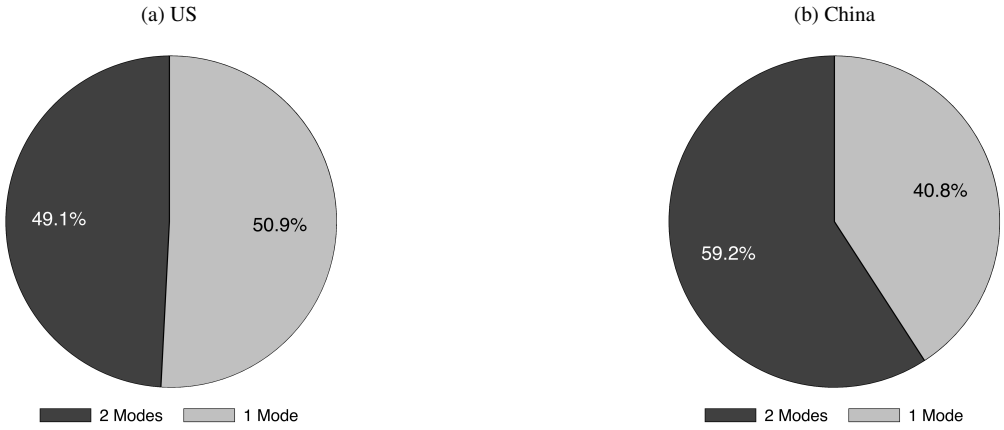
Notes: Panel (a) shows the percentage of origin-destination-HS6 sectors that use both air and ocean transport unweighted. Panel (b) shows the percentage of origin-destination-HS6 sectors that use both air and ocean transport weighted with kilos. The sample consists of imports and exports of European Union members in 2019, collected from Eurostat.

Figure H.4: Number of Modes Used per Origin-Destination-HS6: Imports vs Exports



Notes: Panel (a) shows the percentage of origin-destination-HS6 sectors that use both air and ocean transport for imports. Panel (b) shows the percentage of origin-destination-HS6 sectors that use both air and ocean transport for exports. The sample consists of imports and exports of European Union members in 2019, collected from Eurostat.

Figure H.5: Number of Modes Used per Origin-Destination-HS6: US vs China



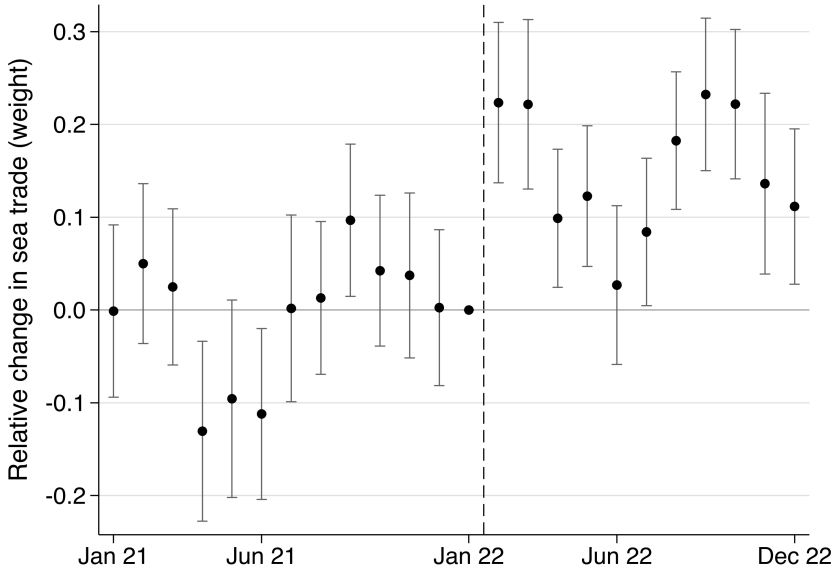
Notes: Panel (a) shows the percentage of origin-destination-HS6 sectors that use both air and ocean transport from the United States. Panel (b) shows the percentage of origin-destination-HS6 sectors that use both air and ocean transport from China. The sample consists of imports and exports of European Union members in 2019, collected from Eurostat.

Figure H.6: Flight Path Cargo vs Passenger



Notes: Flight route between Frankfurt and Beijing after the start of the Ukraine war for a passenger flight vs a cargo flight. Author's own calculation based on Flightradar24 data.

Figure H.7: Sea Trade relative to Air Trade (Weight)



Notes: This figure reports the event study based on equation (17) and shows that there is a significant increase in sea trade, in kilograms, relative to air trade after the closure of the Russian and Ukrainian airspace between treated and non-treated country pairs.

H.1 Additional Tables

Table H.13: Air Trade and Product Characteristics

	% with two modes	Value by air (%)	Weight by air (%)	Observations
Panel A. Broad Economic Classification (BEC)				
Other (0)	35.95	0.021	0.005	7,710
Capital (1)	53.79	0.390	0.060	377,449
Intermediate (2)	54.38	0.271	0.004	1,209,396
Consumption (3)	49.30	0.274	0.025	627,600
Panel B. Containerizability				
Not suitable (0)	52.99	0.054	0.001	193,513
Slightly suitable (1)	44.35	0.202	0.010	151,770
Suitable (2)	53.44	0.347	0.018	1,876,892
Panel C. Temperature Sensitivity				
Low (1)	53.61	0.242	0.005	1,488,940
Medium–Low (2)	54.63	0.285	0.006	433,532
Medium–High (3)	48.78	0.515	0.015	178,331
High (4)	41.85	0.332	0.024	121,372
Panel D. Unit Value				
Low (1)	46.41	0.014	0.002	424,815
Medium (2)	57.46	0.103	0.025	785,724
High (3)	51.83	0.445	0.098	1,011,604
Panel E. Time Costs				
Low (1)	47.56	0.405	0.011	653,082
Medium (2)	54.17	0.326	0.027	814,913
High (3)	55.80	0.085	0.003	754,180

Notes: This table reports descriptive statistics for Eurostat trade flows by product characteristics. The Broad Economic Classification (BEC) categorizes products by final use. Containerizability measures the ease with which a good can be transported in containers (from Bernhofen et al., 2016). Temperature sensitivity is an index ranging from 1 (low) to 4 (high). Unit value is defined as terciles of the sector-level unit value distribution in 2019. Time costs are based on a discretized version of the continuous index in Hummels and Schaur (2013).

Table H.14: Substitution Between Modes of Transport: Weight

	(1) ln FlightTime	(2) $\ln\left(X^{Sea}/X^{Air}\right)$	(3) $\ln\left(X^{Sea}/X^{Air}\right)$	(4) $\ln\left(X^{Sea}/X^{Air}\right)$
ln FlightTime			1.895	2.267
			(0.611)	(0.785)
Post × Treated	0.071	0.134		
	(0.015)	(0.030)		
ln Freight				0.039
				(0.023)
$\hat{\eta}$			3	3.27
Obs.	2,822,346	2,822,346	2,822,346	2,822,346
Origin-Destination-HS6	✓	✓	✓	✓
Year-Month-HS6	✓	✓	✓	✓
F-Test			23.15	18.39
R^2	0.85	0.59	-0.01	-0.02

Notes:Column (1) reports estimates from regression (16), where the dependent variable is the log average flight time for an origin–destination pair. Column (2) reports estimates from regression (17), where the dependent variable is the log ratio (in kilograns) of sea to air trade between origin country i and destination country n in sector h and month t. Treated equals one for origin–destination pairs involving Asia and zero otherwise. Post equals one after the start of the war (February 2022) and zero otherwise. Columns (3)–(4) estimate the Wald-DiD elasticity in equation (18) via 2SLS; $\hat{\eta}+1$ is the implied elasticity of substitution between air and sea transport. Standard errors are clustered at the origin–destination–year level. Column (4) additionally controls for freight indices from major European ports to Asian and American destinations.

Table H.15: Substitution Between Modes of Transport:: UV ratio

	(1) Low	(2) Med-low	(3) Med-High	(4) High
lnftime	2.533	1.807	1.805	1.051
	(1.120)	(0.709)	(0.585)	(0.497)
$\hat{\eta}$	3.53	2.81	2.80	2.05
Obs.	282,204	675,478	924,758	939,399
Origin-Destination-HS6	✓	✓	✓	✓
Year-Month-HS6	✓	✓	✓	✓
F-Test	11.43	20.25	25.48	27.67

Notes: The dependent variable is the log average flight time between two country pairs. The sample is divided based on the ratio of the unit values by mode within an origin-destination-sector. Standard errors are clustered at the origin-destination-year level.

Table H.16: Impact on Within Product Composition

	(1) UV Ratio	(2) UV Air	(3) UV Sea
ln FlightTime	0.195	0.195	-0.000
	(0.221)	(0.233)	(0.222)
Obs.	3,069,425	3,069,425	3,069,425
Origin-Destination-HS6	✓	✓	✓
Year-Month-HS6	✓	✓	✓
F-Test	23.72	17.64	17.64

Notes: This table reports the estimates for the Wald-Did estimator computed via IV-2SLS by unit values calculated at the origin-destination-sector-transport level. Column (1) reports the results for the ratio between air and sea unit values within a common origin-destination-sector triplet. Columns (2) and (3) instead estimate the effect separately. Standard errors are clustered at the origin-destination level.

Table H.17: Substitution Between Modes of Transport: Naive OLS

	(1) $\ln\left(X^{Sea}/X^{Air}\right)$	(2) $\ln\left(X^{Sea}/X^{Air}\right)$
ln FlightTime	-0.054	-0.042
	(0.050)	(0.048)
Obs.	2,821,847	2,753,888
Origin-Destination-HS6	✓	✓
Year-Month-HS6	✓	✓
R^2	0.61	0.58

Notes: This table reports the estimates for "naive" OLS in which I regress changes in the relative trade share by mode on flight time. Standard errors are clustered at the origin-destination level.

Table H.18: Comparison of Treated and Control Groups: U.S. Census Data

	Group	Mean	Median	Std. Dev.
Panel A. Air Trade				
Log value	Non-treated	11.48	11.41	2.26
	Treated	11.86	11.85	1.45
Log cost	Non-treated	1.19	1.29	1.25
	Treated	1.34	1.50	1.22
Panel B. Sea Trade				
Log value	Non-treated	12.07	12.15	2.28
	Treated	13.53	13.68	2.44
Log cost	Non-treated	-0.67	-0.59	1.47
	Treated	-0.63	-0.60	1.17

Notes: The table compares treated and control groups to assess whether the estimated effects are driven by pre-shock differences in trade composition. The treated group consists of U.S. trading partners in Asia, while the control group includes European partners. The unit of observation is monthly bilateral trade flows by transport mode at the country-pair–sector level. Sectors are defined at the HS6 level. Data refer to 2019, prior to the shock and the COVID-19 pandemic.

Table H.19: List of Entities Used

Code	Entity	Code	Entity
ARE	United Arab Emirates	MEX	Mexico
ARG	Argentina	MMR	Myanmar
AUS	Australia	MUS	Mauritius
BIH	Bosnia and Herzegovina	MYS	Malaysia
BLR	Belarus	NIC	Nicaragua
BRA	Brazil	NZL	New Zealand
CAN	Canada	OMN	Oman
CHN	China[a]	PER	Peru
COL	Colombia	PHL	Philippines
CRI	Costa Rica	QAT	Qatar
ECU	Ecuador	RUS	Russian Federation
EGY	Egypt	SAU	Saudi Arabia
EUR	European bloc[b]	SEN	Senegal
GBR	United Kingdom	SGP	Singapore
GEO	Georgia	SRB	Serbia
IDN	Indonesia	TJK	Tajikistan
IND	India	TUN	Tunisia
IRN	Iran	TUR	Türkiye
JPN	Japan	TZA	Tanzania
KEN	Kenya	UKR	Ukraine
KOR	Korea	URY	Uruguay
KWT	Kuwait	USA	United States
LKA	Sri Lanka	VNM	Viet Nam
MAR	Morocco	ZAF	South Africa
MDA	Moldova	ROW	Rest of the world

Notes: The calibration has $N = 50$ entities (Appendix G.3). [a]A China bloc (China, Hong Kong, and Macao), referred to as China. [b]The 27 members of the European Union, Iceland, Norway, and Switzerland. The rest of the world (ROW) contains all other countries in BACI. Trade among members of a bloc is treated as domestic.

Table H.20: Welfare Effects of Airspace Closure

Entity	Sea to air import ratio	Welfare change			Difference from baseline	
		Baseline	No Subst.	No Modes	No Subst.	No Modes
Argentina	+0.16	−0.005	−0.005	−0.011		
Australia	+0.21	−0.006	−0.006	−0.027		
Belarus	+0.13	−0.011	−0.011	−0.034		
Bosnia and Herzegovina	+0.26	0.019	0.021	0.022		
Brazil	+0.18	−0.004	−0.004	−0.013		
Canada	+0.08	−0.005	−0.005	−0.017		
China	+4.32	−0.026	−0.029	−0.123	+9.34	+369.17
Colombia	+0.07	−0.005	−0.005	−0.020		
Costa Rica	−0.15	−0.007	−0.008	−0.027		
Ecuador	+0.03	−0.012	−0.013	−0.036		
Egypt	+0.30	−0.003	−0.003	−0.018		
European bloc	+5.85	−0.042	−0.046	−0.155	+9.45	+268.68
Georgia	+0.15	−0.012	−0.012	−0.063		
India	+0.13	−0.003	−0.002	−0.006		
Indonesia	+1.09	−0.025	−0.027	−0.104	+6.92	+317.90
Iran	+0.21	−0.002	−0.002	−0.014		
Japan	+3.16	−0.027	−0.029	−0.071	+5.50	+160.03
Kenya	−0.01	−0.008	−0.008	−0.027		
Korea	+2.73	−0.039	−0.041	−0.119	+6.39	+209.74
Kuwait	+0.05	−0.011	−0.012	−0.048		
Malaysia	+1.60	−0.155	−0.157	−0.263	+1.06	+69.75
Mauritius	−0.09	0.005	0.006	−0.017		
Mexico	+0.13	−0.003	−0.003	−0.008		
Moldova	+0.04	0.028	0.029	0.003		
Morocco	+0.20	0.025	0.028	0.020		
Myanmar	+0.03	−0.048	−0.057	−0.402	+19.28	+743.37
New Zealand	+0.12	−0.008	−0.009	−0.044		
Nicaragua	+0.06	−0.011	−0.012	−0.039		
Oman	+0.04	−0.007	−0.007	−0.040		
Peru	+0.09	−0.006	−0.006	−0.017		
Philippines	+1.29	−0.054	−0.056	−0.227	+5.49	+323.80
Qatar	+0.20	−0.006	−0.007	−0.076		
Russian Federation	+0.13	−0.009	−0.010	−0.031		
Saudi Arabia	+0.14	−0.002	−0.002	−0.018		
Senegal	+0.01	−0.014	−0.015	−0.068		
Serbia	+0.25	0.006	0.006	−0.027		
Singapore	+2.89	−0.157	−0.165	−0.553	+5.02	+251.34
South Africa	+0.12	−0.006	−0.006	−0.011		
Sri Lanka	+0.09	0.002	0.004	0.015		
Türkiye	+0.19	−0.001	0.000	0.003		
Tajikistan	−0.00	−0.017	−0.017	−0.017		
Tanzania	+0.02	−0.008	−0.009	−0.024		
Tunisia	+0.22	0.030	0.034	0.030		
Ukraine	+0.16	−0.009	−0.011	−0.056		
United Arab Emirates	+0.11	−0.012	−0.013	−0.057		
United Kingdom	+0.27	−0.002	−0.003	−0.017		
United States	+0.23	−0.002	−0.002	−0.009		
Uruguay	+0.01	−0.009	−0.010	−0.022		
Viet Nam	+0.83	−0.093	−0.102	−0.463	+9.31	+396.69
Rest of the world	+0.13	−0.003	−0.003	−0.016		

Notes: All values are in percent. *Sea to air import ratio* is the change in the ratio of sea to air imports of the entity, aggregated over partners and sectors, in the baseline model. *Welfare change* is relative to the initial equilibrium under the baseline model, the model without substitution across modes (Cobb–Douglas modal shares), and the model without transport modes. *Difference from baseline* is the percent difference between the welfare change of the restricted model and that of the baseline; positive values indicate a larger welfare change in absolute value, that is, a larger loss for entities that lose. The shock is a permanent 12 percent increase in air transport costs on routes between EUR and China, Indonesia, Japan, Korea, Malaysia, Myanmar, the Philippines, Singapore, and Viet Nam. Differences from the baseline are reported for these ten directly affected entities only. Parameters: η_j as in Table G.12, $\lambda_{\text{air}} = 0.097$, $\lambda_{\text{sea}} = 0.06$.

Table H.21: Welfare Effects of Suez Canal Closure

	Sea to air	Welfare change			Difference from baseline	
Entity	import ratio	Baseline	No Subst.	No Modes	No Subst.	No Modes
Argentina	+1.33	−0.006	−0.008	−0.024	+31.97	+288.41
Australia	+1.89	−0.339	−0.345	−0.434	+1.61	+28.08
Belarus	+0.50	−0.102	−0.105	−0.123	+3.05	+20.34
Bosnia and Herzegovina	+0.62	0.097	0.133	0.232	+36.80	+139.99
Brazil	+2.11	−0.042	−0.044	−0.054	+4.89	+29.55
Canada	−1.15	−0.104	−0.107	−0.135	+2.48	+28.82
China	−14.46	−0.347	−0.359	−0.427	+3.50	+23.00
Colombia	−1.81	−0.131	−0.134	−0.162	+2.12	+23.55
Costa Rica	+15.81	−0.156	−0.161	−0.195	+3.63	+25.40
Ecuador	−1.57	−0.221	−0.229	−0.293	+3.61	+32.66
Egypt	−0.66	−0.036	−0.034	−0.043	−3.66	+22.10
European bloc	−26.13	−0.450	−0.479	−0.625	+6.37	+38.79
Georgia	−10.89	−2.454	−2.633	−3.293	+7.26	+34.18
India	−17.36	−0.652	−0.702	−1.066	+7.74	+63.59
Indonesia	−5.82	−0.399	−0.411	−0.506	+2.83	+26.57
Iran	−17.32	−1.007	−1.046	−1.163	+3.83	+15.49
Japan	−4.46	−0.119	−0.122	−0.177	+2.45	+48.77
Kenya	−9.72	−0.882	−0.904	−1.031	+2.47	+16.81
Korea	−8.24	−0.275	−0.286	−0.369	+3.71	+33.95
Kuwait	−26.17	−2.876	−2.897	−3.034	+0.71	+5.49
Malaysia	−9.52	−0.737	−0.757	−1.458	+2.75	+98.01
Mauritius	−18.37	−2.813	−2.796	−2.744	−0.59	−2.46
Mexico	+0.24	−0.047	−0.049	−0.065	+2.82	+37.48
Moldova	−1.03	−0.051	−0.030	0.036	−41.82	−170.01
Morocco	−4.54	−0.660	−0.653	−0.847	−1.16	+28.27
Myanmar	−6.33	−2.024	−2.116	−2.285	+4.58	+12.93
New Zealand	−1.47	−0.314	−0.321	−0.397	+2.32	+26.50
Nicaragua	−0.26	−0.424	−0.430	−0.487	+1.42	+15.00
Oman	−11.06	−2.120	−2.198	−2.522	+3.70	+18.98
Peru	+0.20	−0.089	−0.090	−0.084	+0.86	−5.95
Philippines	−3.39	−0.811	−0.831	−0.988	+2.42	+21.82
Qatar	−20.90	−3.427	−3.555	−3.934	+3.74	+14.81
Russian Federation	−11.20	−0.500	−0.517	−0.648	+3.25	+29.53
Saudi Arabia	−26.29	−1.918	−2.027	−2.283	+5.67	+19.03
Senegal	−11.36	−1.126	−1.141	−1.387	+1.33	+23.15
Serbia	−1.80	−0.094	−0.094	−0.084	+0.58	−10.10
Singapore	−13.35	−2.442	−2.515	−3.064	+2.98	+25.48
South Africa	+0.04	−0.085	−0.085	−0.088	+0.15	+3.32
Sri Lanka	−9.18	−1.102	−1.199	−1.535	+8.82	+39.34
Türkiye	−22.78	−1.073	−1.123	−1.398	+4.63	+30.24
Tajikistan	−0.96	−0.195	−0.190	−0.034	−2.40	−82.77
Tanzania	−6.99	−0.725	−0.750	−0.864	+3.46	+19.12
Tunisia	−6.04	−1.228	−1.252	−1.528	+1.98	+24.44
Ukraine	−5.15	−1.394	−1.481	−1.640	+6.20	+17.64
United Arab Emirates	−14.65	−3.277	−3.474	−5.265	+6.02	+60.67
United Kingdom	−12.50	−0.481	−0.500	−0.648	+3.91	+34.61
United States	−2.05	−0.076	−0.078	−0.109	+2.47	+43.45
Uruguay	+1.56	−0.047	−0.050	−0.075	+6.88	+61.05
Viet Nam	−5.87	−1.861	−1.925	−2.274	+3.44	+22.20
Rest of the world	−26.27	−1.124	−1.143	−1.473	+1.76	+31.10

Notes: All values are in percent. *Sea to air import ratio* is the change in the ratio of sea to air imports of the entity, aggregated over partners and sectors, in the baseline model. *Welfare change* is relative to the initial equilibrium under the baseline model, the model without substitution across modes (Cobb–Douglas modal shares), and the model without transport modes. *Difference from baseline* is the percent difference between the welfare change of the restricted model and that of the baseline; positive values indicate a larger welfare change in absolute value, that is, a larger loss for entities that lose. The shock raises sea transport costs on each route in proportion to the increase in sea distance from closing the canal Feyrer (2021); all routes to and from Egypt are unaffected. Parameters: η_j as in Table G.12, $\lambda_{\text{air}} = 0.097$, $\lambda_{\text{sea}} = 0.06$.